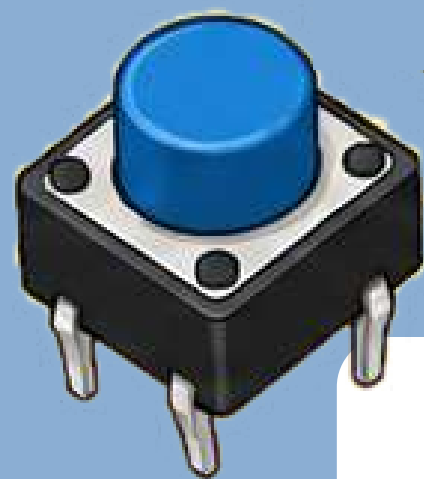
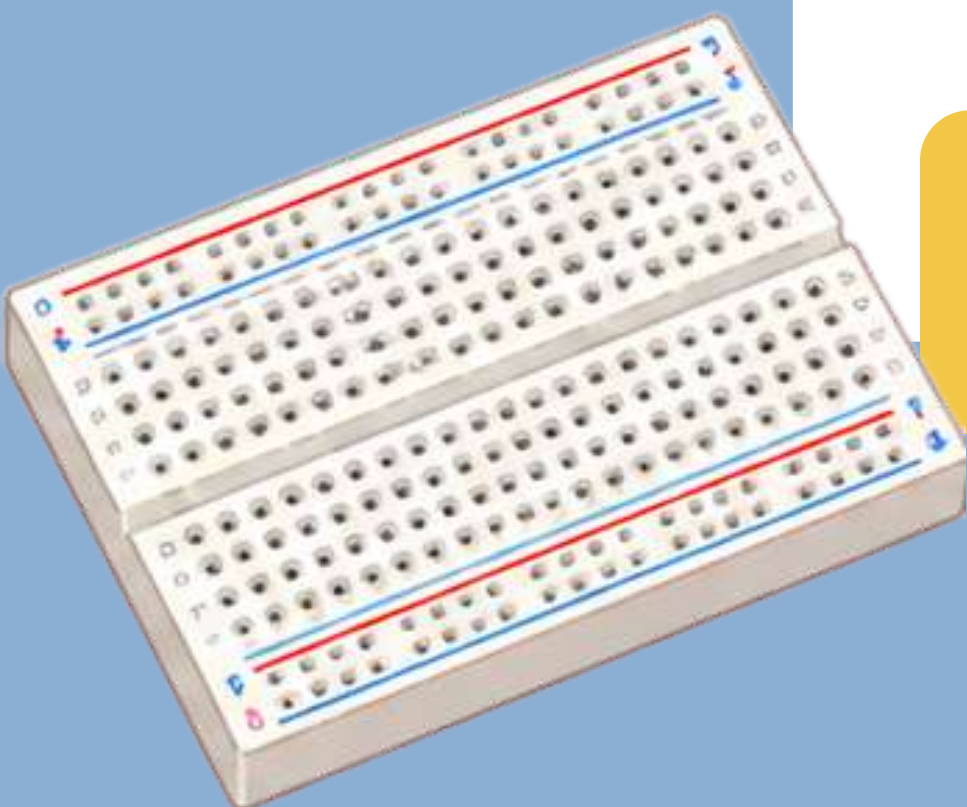




Oficina de FPGA

Conheça o ecossistema do hardware
configurável



**"People who are really serious about software
should make their own hardware."**

— Alan Kay



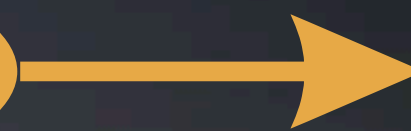
9 de Janeiro de 2007

**"People who are really serious about software
should make their own hardware."**

**Steve
Jobs**



– Alan Kay



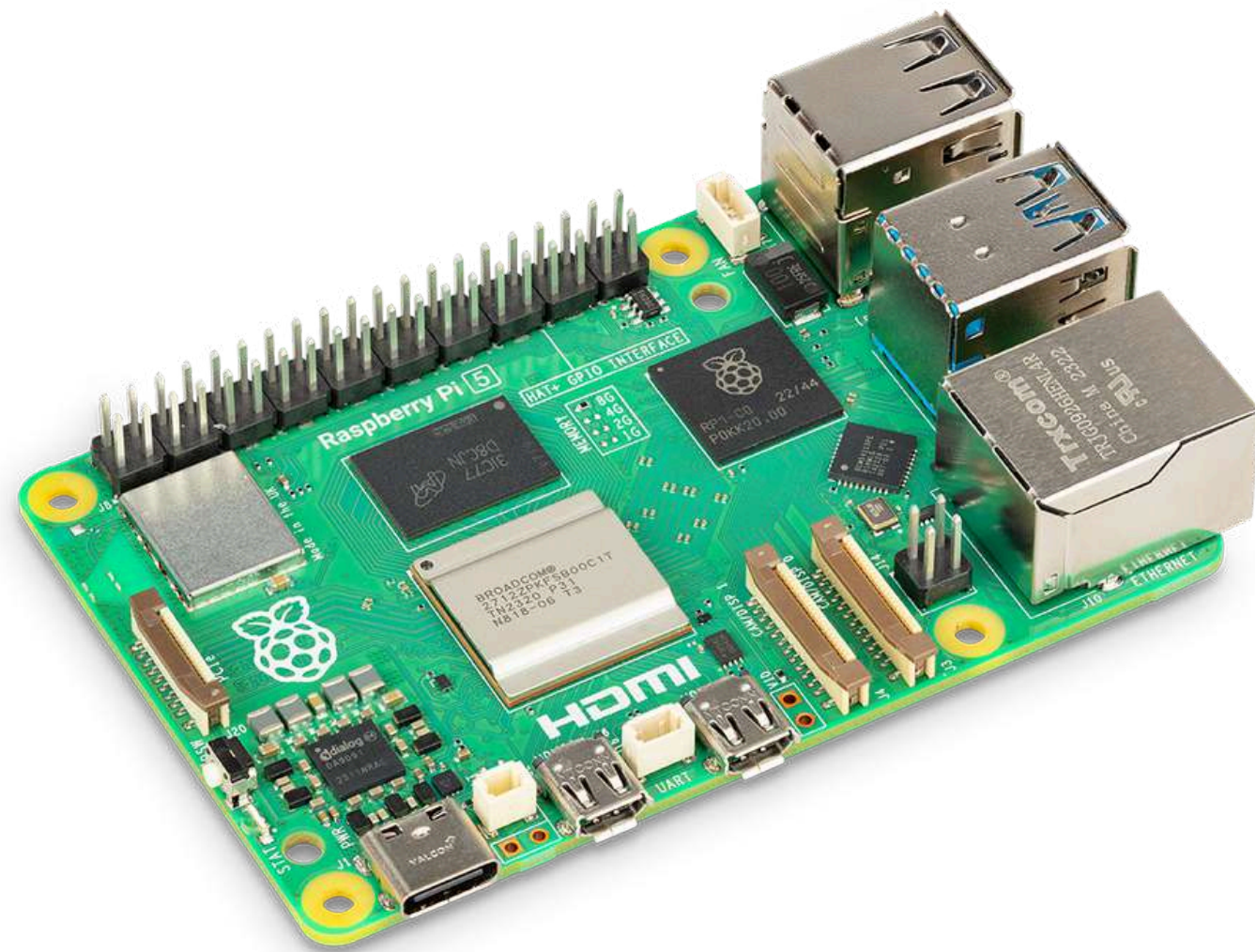
**Prêmio
Turing**

Computadores estão se tornando caixas pretas

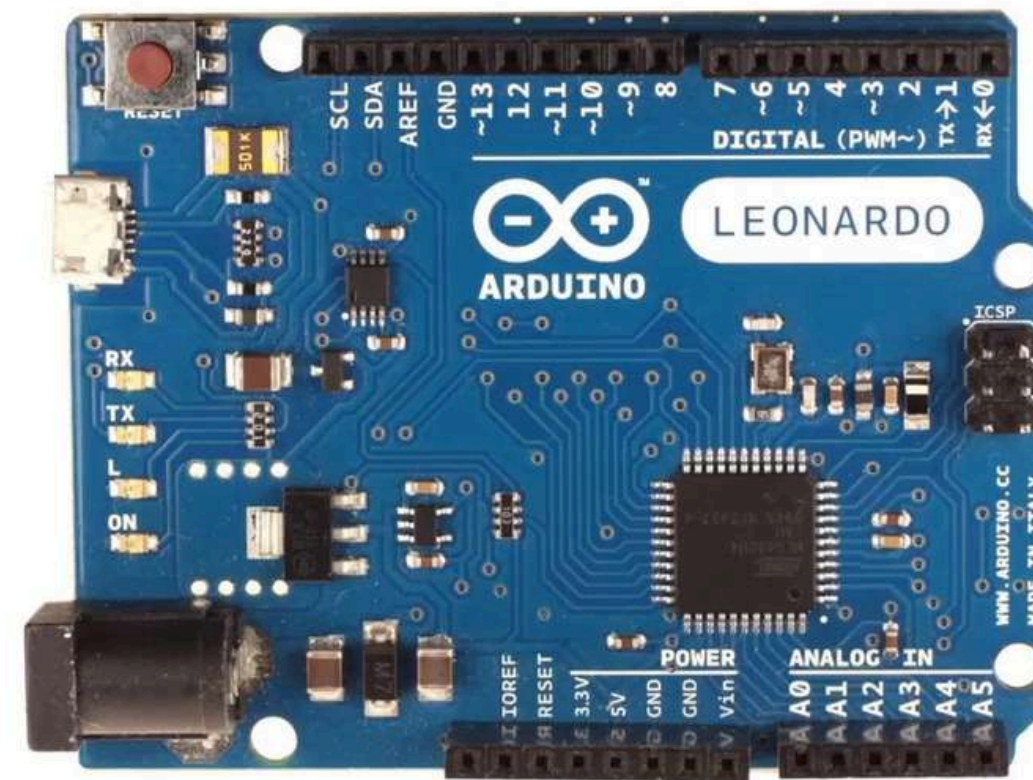
Perdemos o costume de abrir a
carcaça e manipular as peças



Movimento Maker veio Combater esse Fenômeno



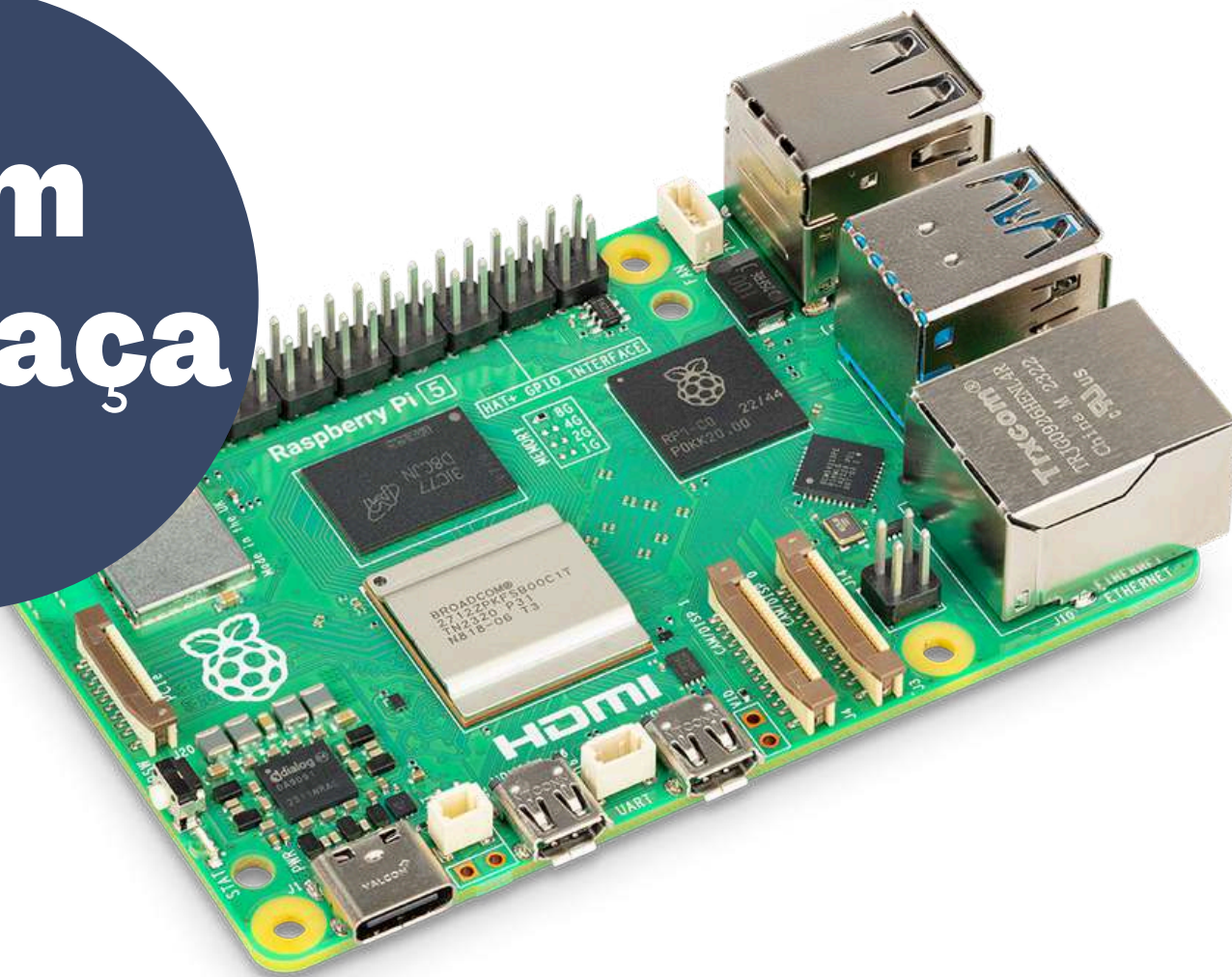
Raspberry Pi



Arduino

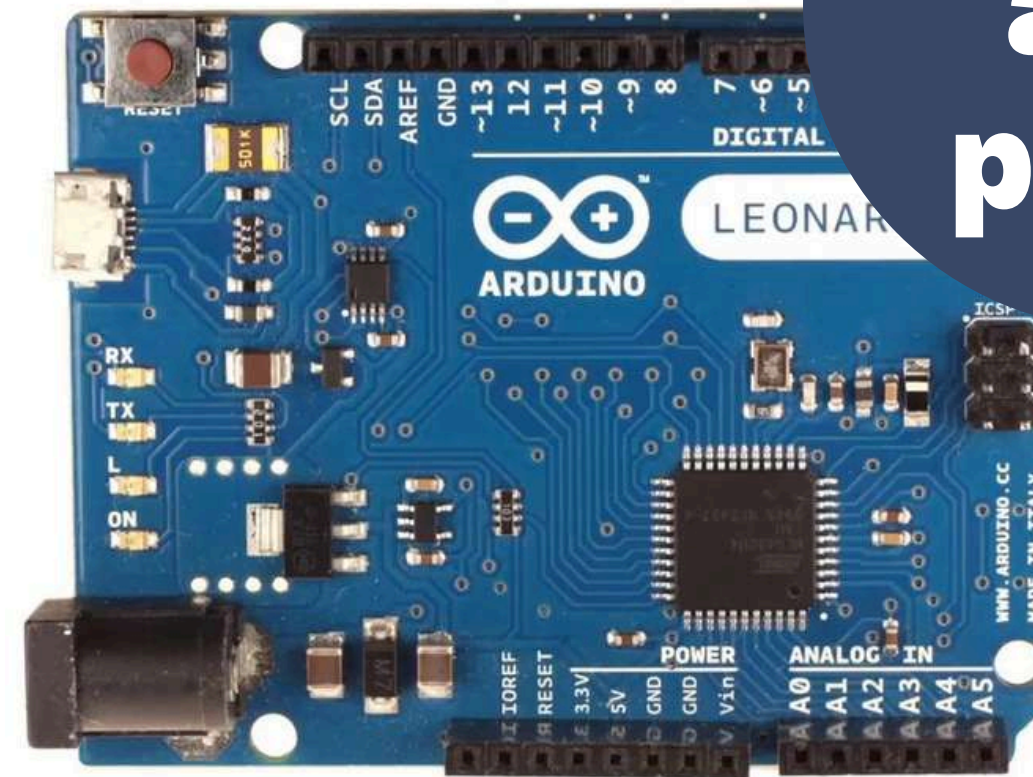
Movimento Maker veio Combater esse Fenômeno

Sem
Carcaca



Raspberry Pi

Acesso
aos
pinos



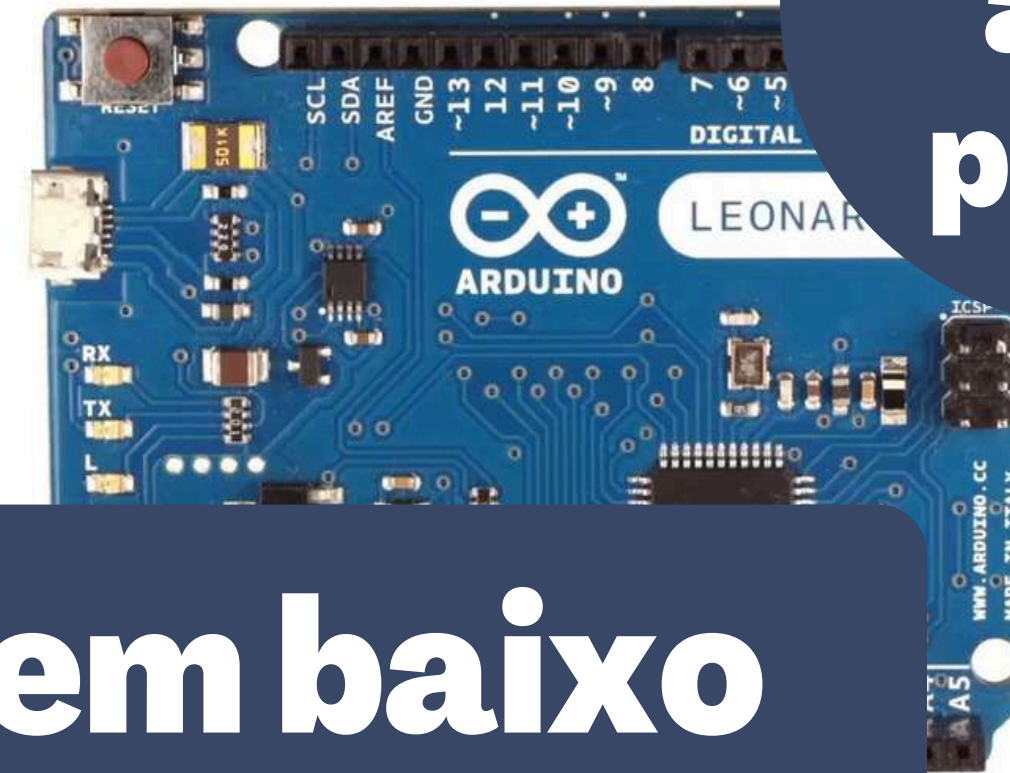
Arduino

Movimento Maker veio Combater esse Fenômeno

Sem
Carcaca



Acesso
aos
pinos

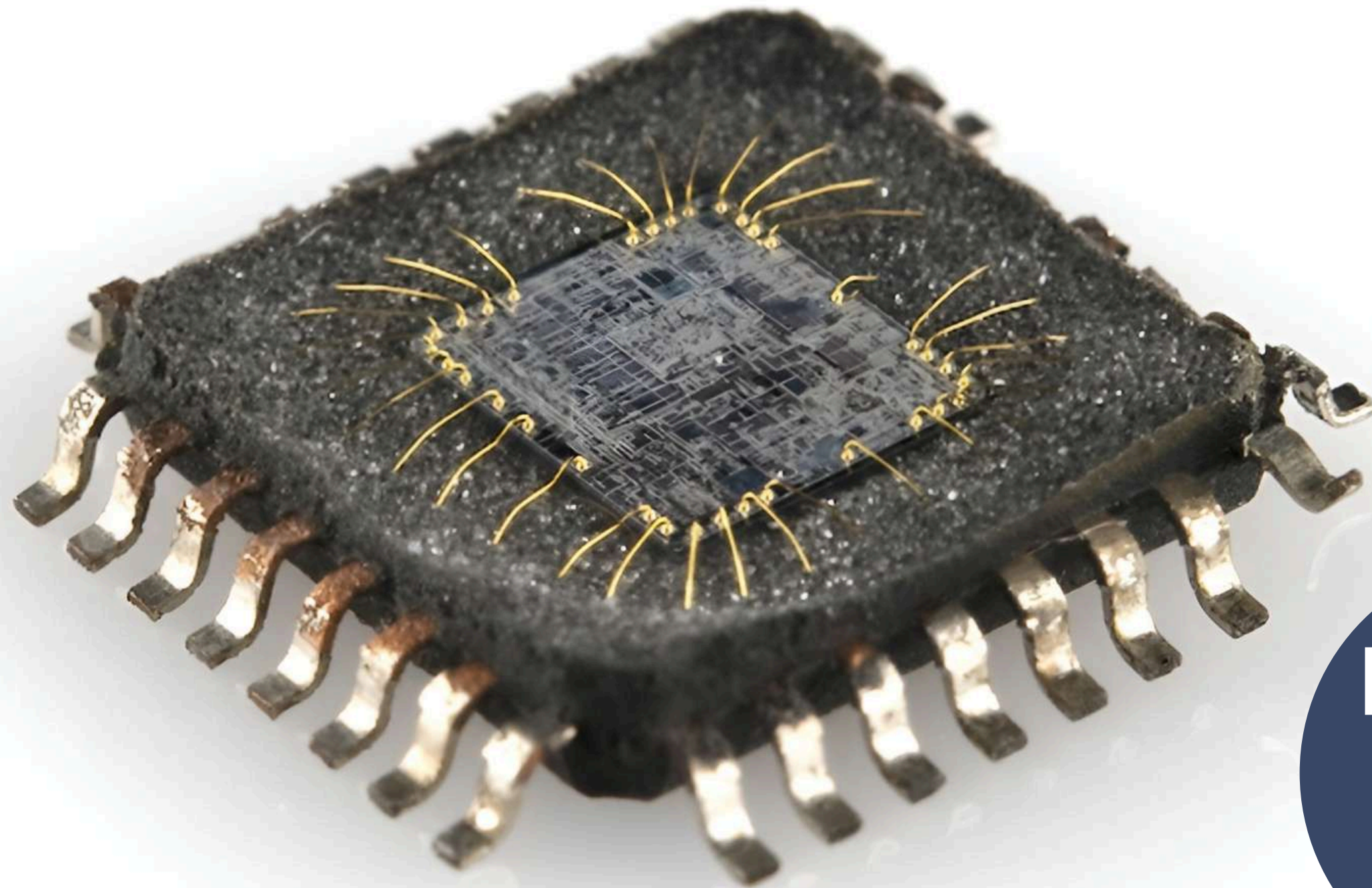


Programação em baixo
nível: protocolos

Raspberry Pi

Arduino

Com FPGAs, damos um passo além



**Dentro
do
Chip**

A stylized illustration of a woman with dark skin and hair in a bun, wearing safety glasses and a white lab coat over a yellow shirt. She is pointing her right index finger towards a yellow box containing the title. The background is a solid blue color.

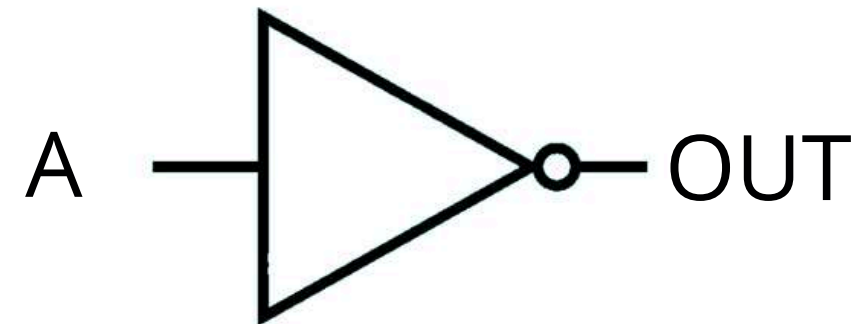
Aprenderemos Hoje

- Funcionamento de um FPGA
- Introdução a HDLs
- Pulse Width Modulation (PWM)
- Código Morse com buzzer



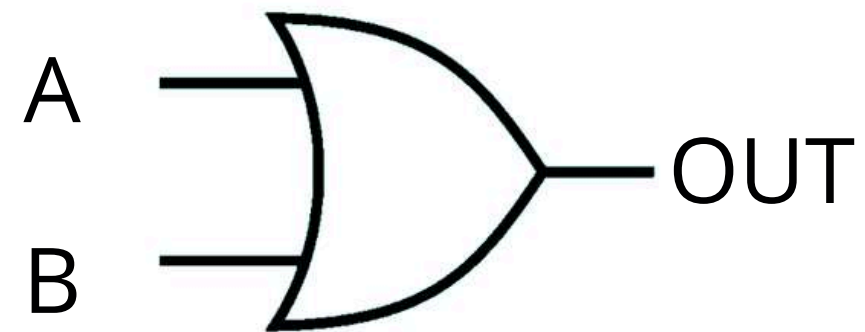
Portas Lógicas

Unidade Fundamental de Sistemas Digitais



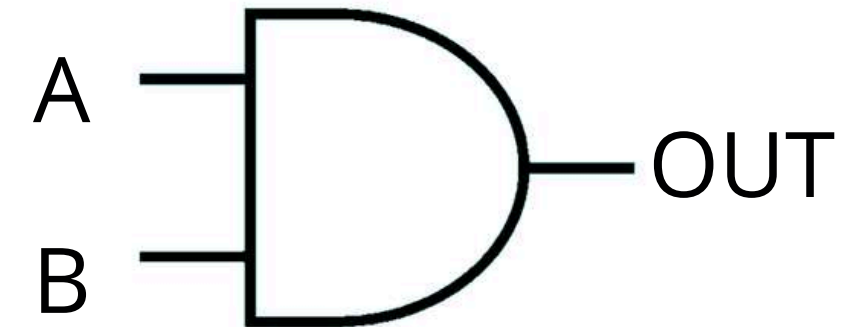
A	OUT
0	1
1	0

NOT



A	B	OUT
0	0	0
0	1	1

OR

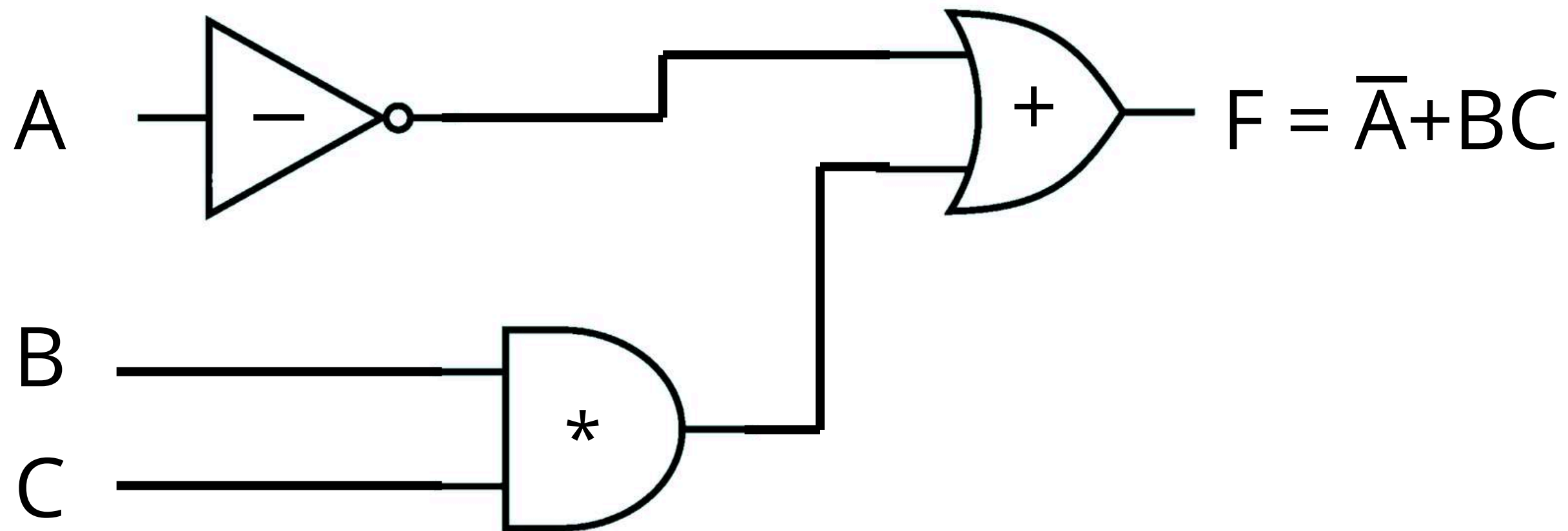


A	B	OUT
0	0	0
1	1	1

AND

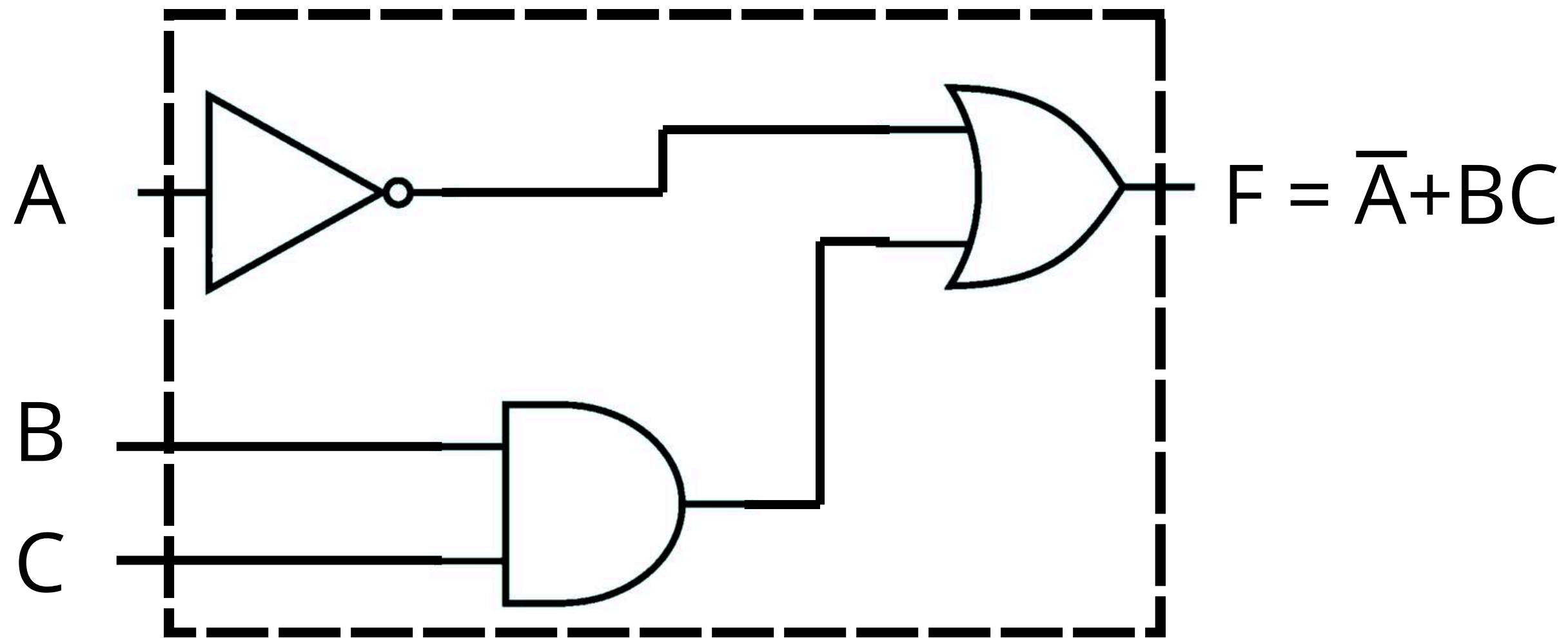
As portas podem ser combinadas

Permite a criação de funções complexas



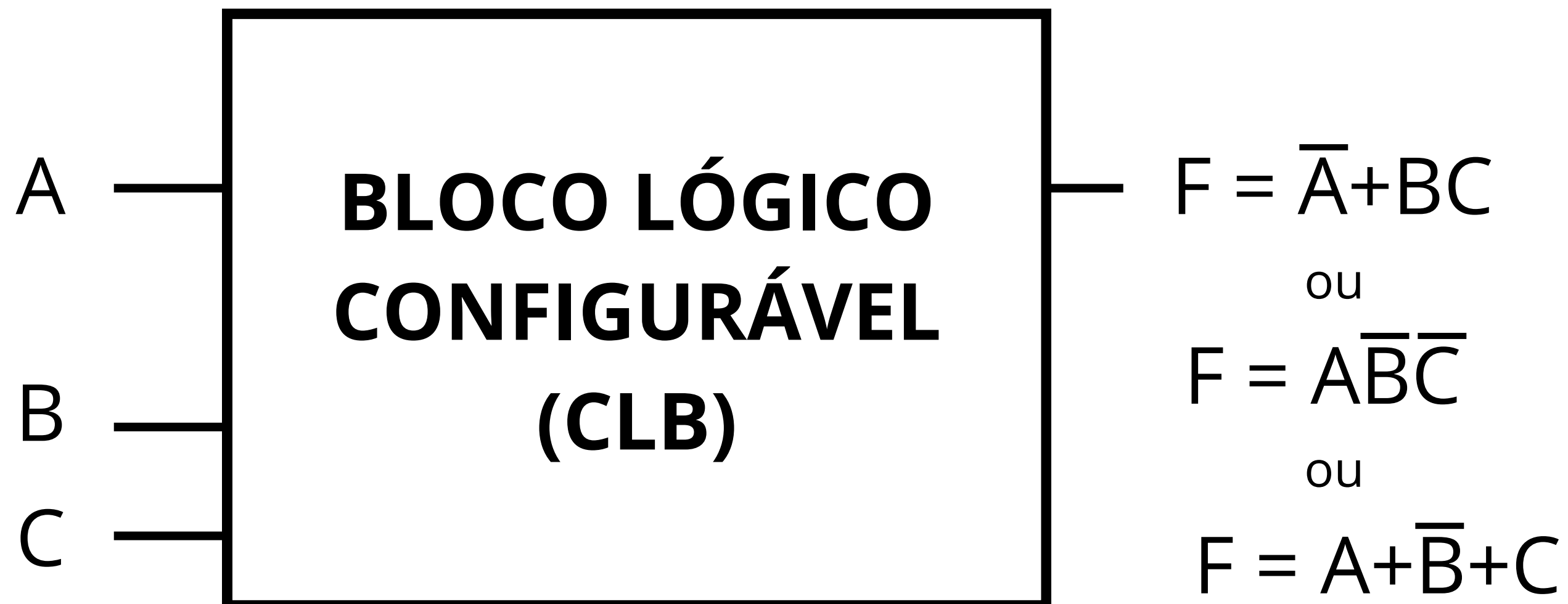
**Imagine um
componente capaz de**

Fazer todo tipo de combinação
lógica



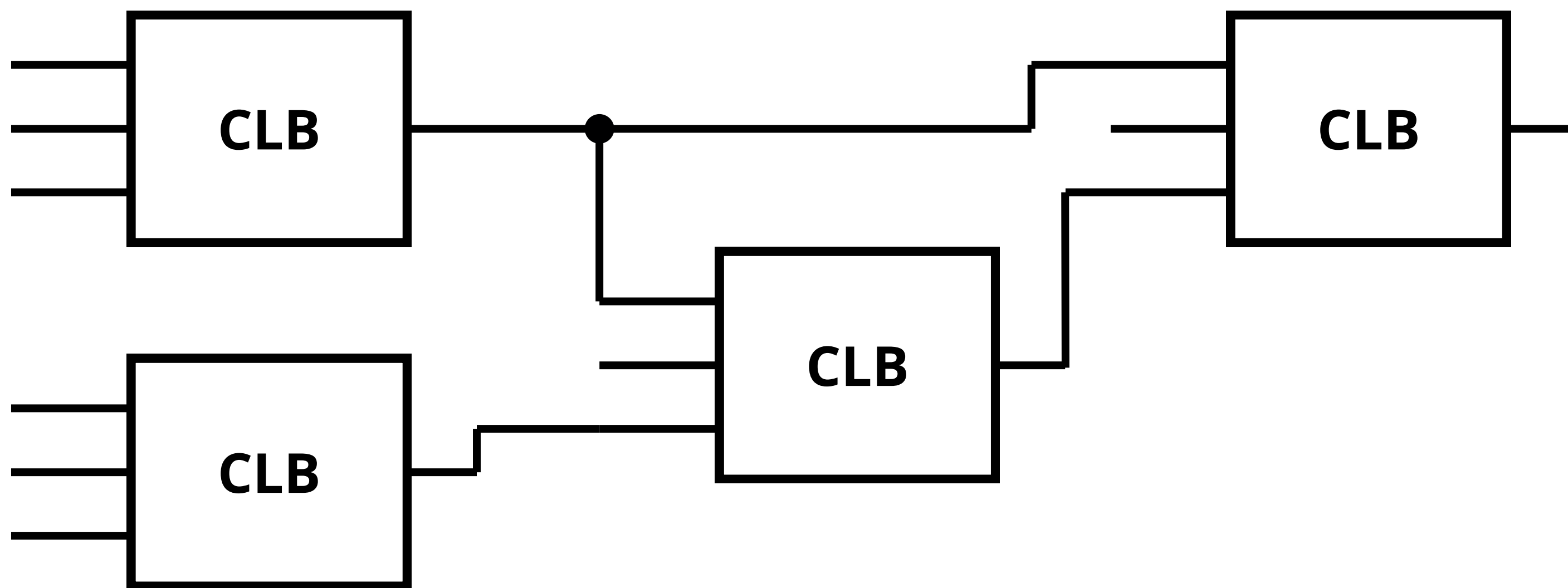
**Agora imagine um
componente capaz de**

Fazer todo tipo de combinação
lógica

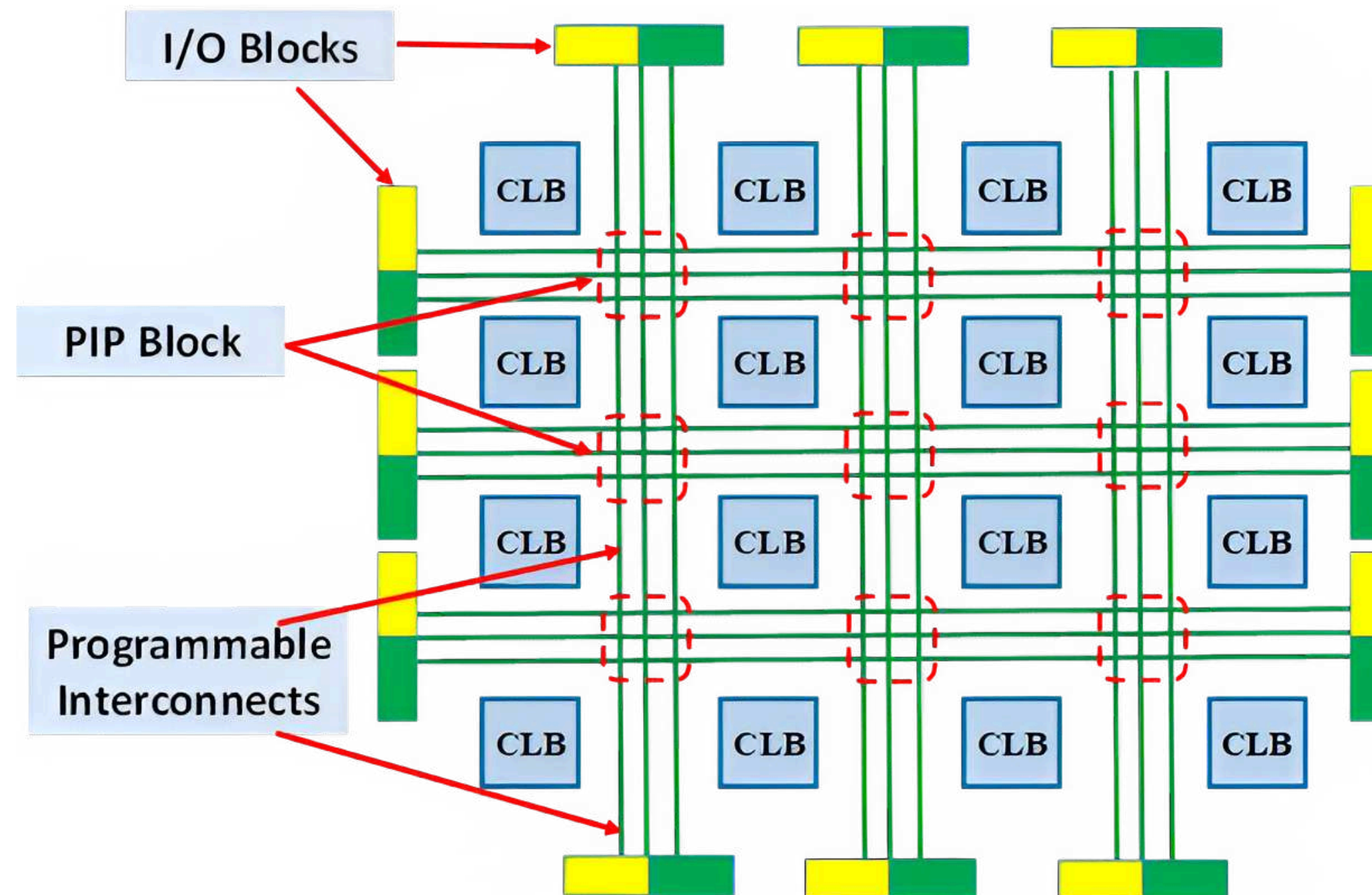


Conecte vários desses blocos

E crie combinações lógicas ainda mais complexas



Um FPGA nada mais é do que uma matriz de CLBs que podem se conectar



Mas, como configurar o FPGA?

Vamos ver agora
Hardware Description
Languages (HDL)

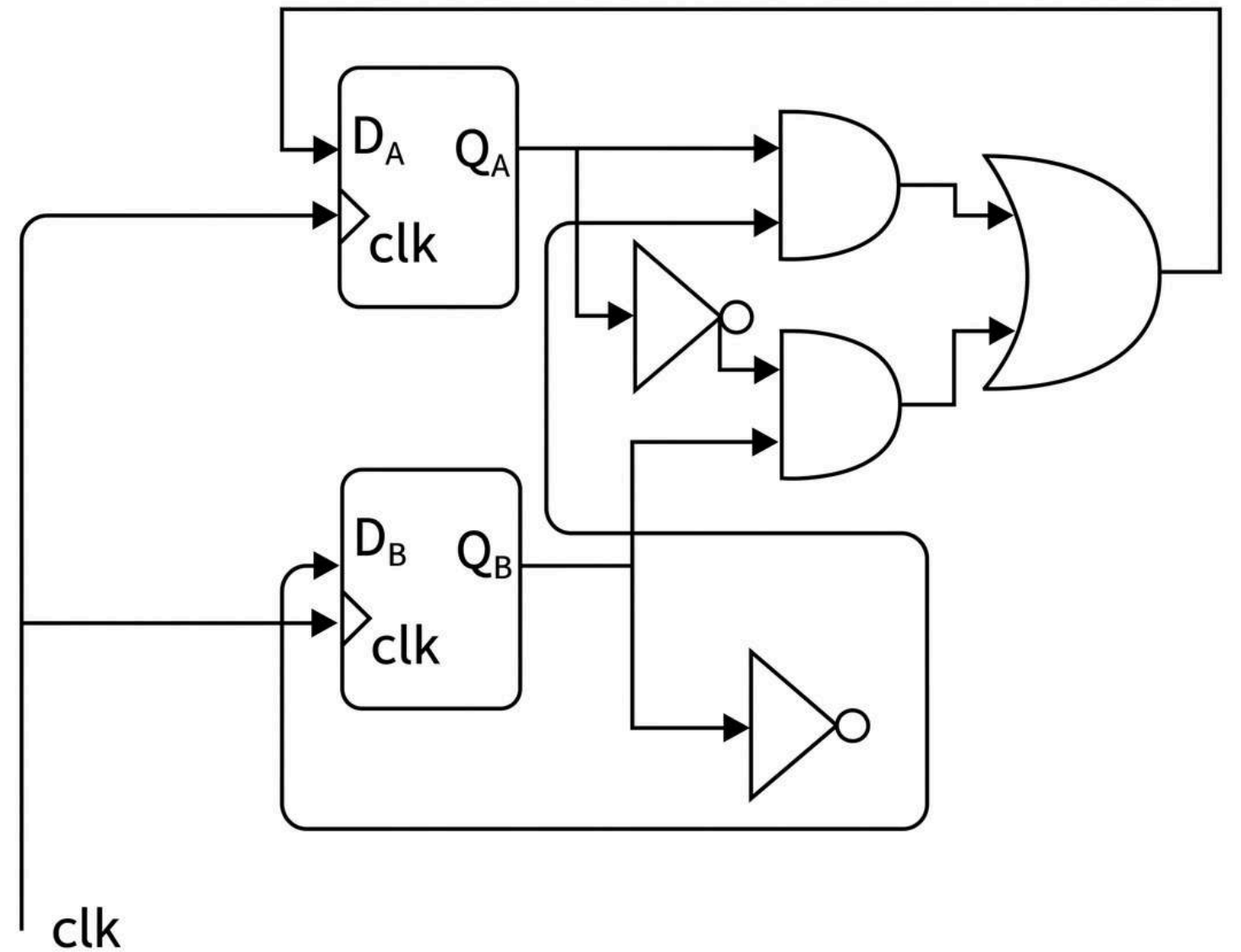


**Com uma linguagem
parecida com as
usadas em software...**

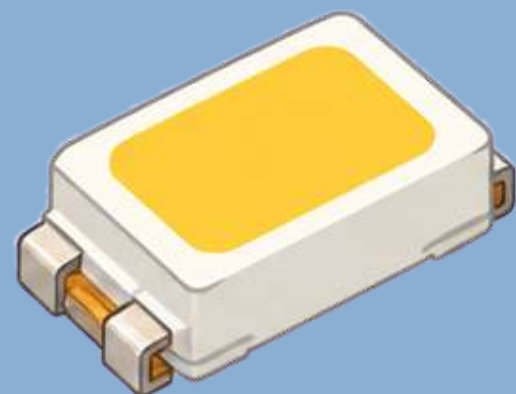
```

1  module EQUALS(
2      input logic [3:0] a,
3      input logic [3:0] b,
4      output logic eq
5  );
6      logic [3:0] tmp;
7
8      generate
9          genvar i;
10         for (i = 0; i < 4; i++) begin :
11             assign tmp[i] = ~(a[i] ^ b[i]);
12         end
13     endgenerate
14
15     assign eq = &tmp;
16
17 endmodule

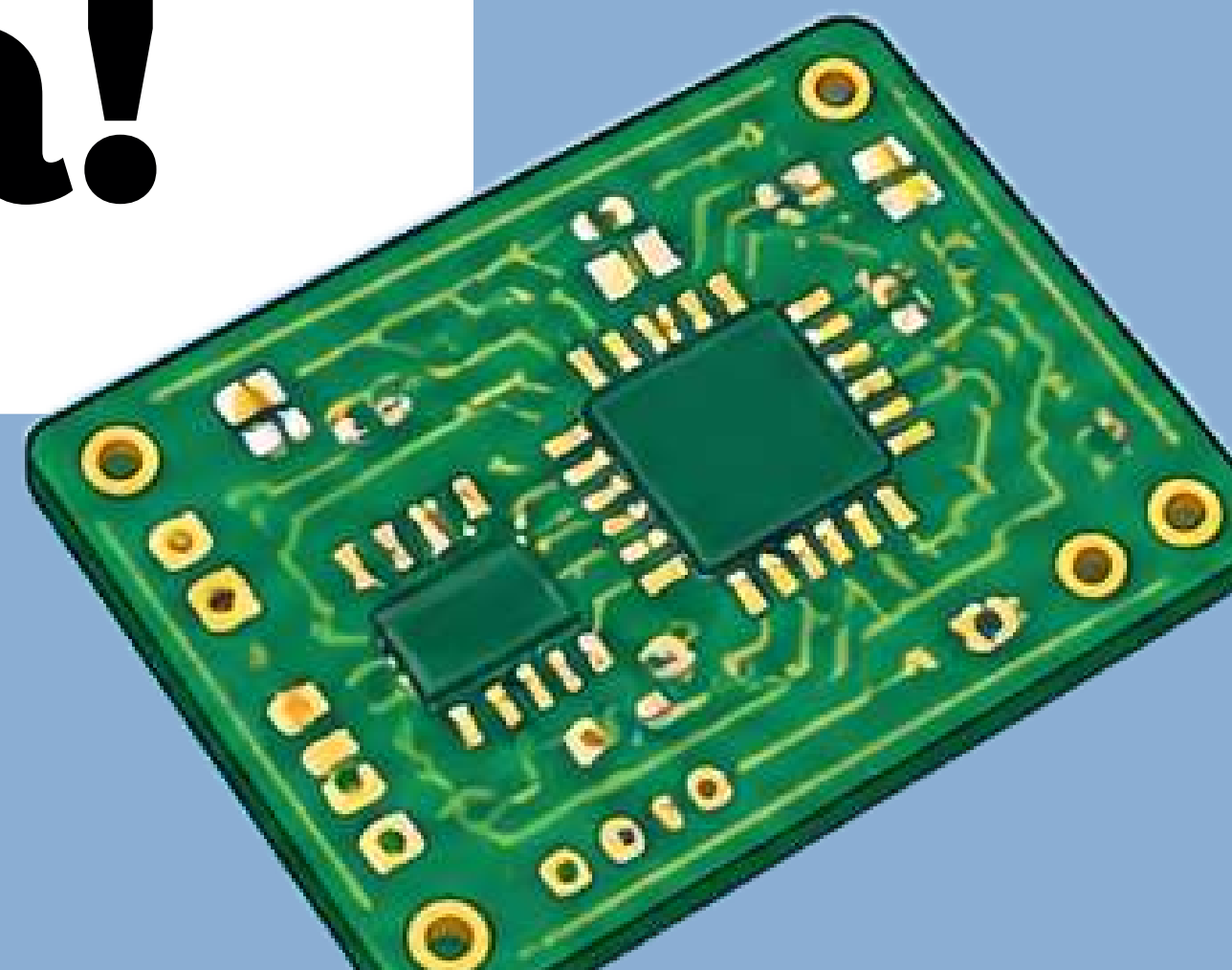
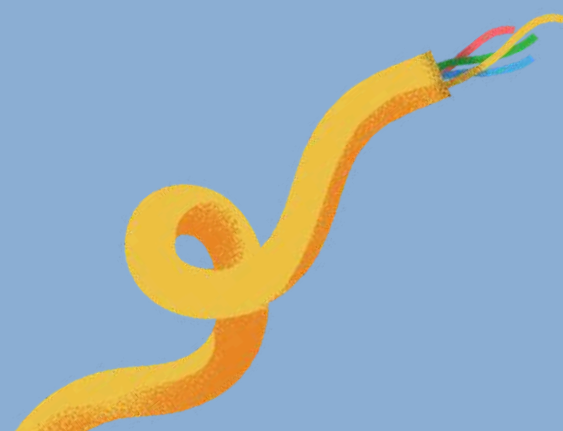
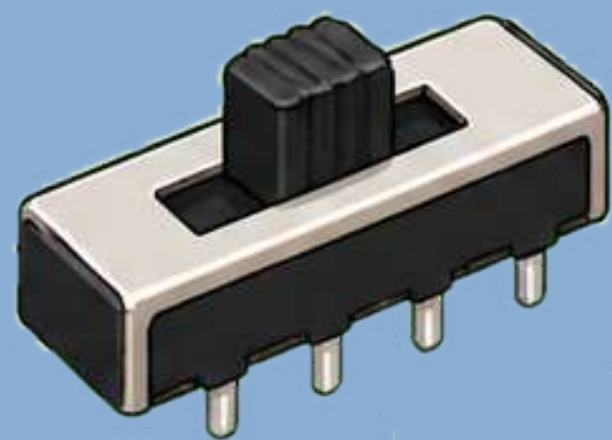
```



...é possível descrever
circuitos digitais arbitrários

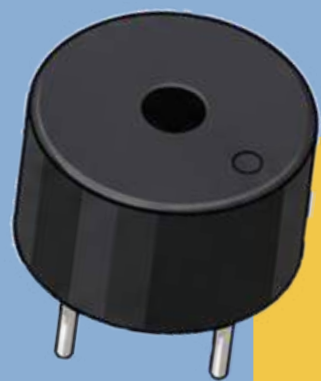


**Vamos para
a prática!**



- Na pasta
"Parte1_PWM"
- Abrir o projeto
"Teste_PWM.qpf" no
Quartus II
- Abrir o arquivo
"Teste_PWM.v"





Interface e Corpo do Módulo

```
1  module EQUALS(  
2      input logic [3:0] a,  
3      input logic [3:0] b,  
4      output logic eq  
5  );  
6      logic [3:0] tmp;  
7  
8      generate  
9          genvar i;  
10         for (i = 0; i < 4; i++) begin :  
11             assign tmp[i] = ~(a[i] ^ b[i]);  
12         end  
13     endgenerate  
14  
15     assign eq = &tmp;  
16  
17 endmodule
```



**Interface define a
conexão com o mundo
externo (pinos)**



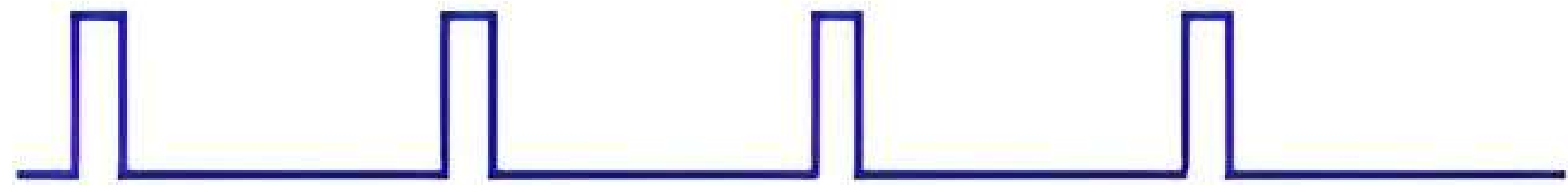
**Corpo define o
comportamento do
componente**



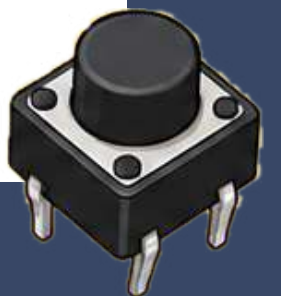
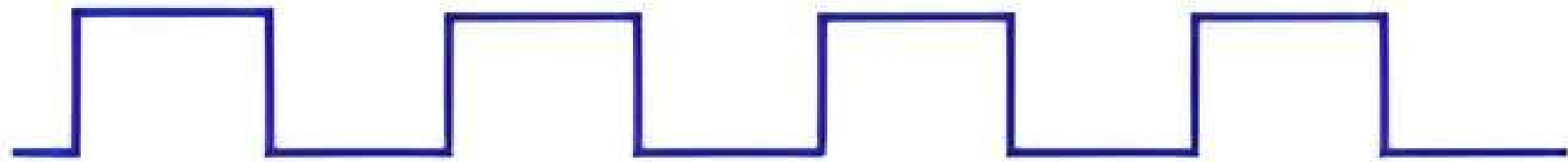
Pulse Width Modulation - PWM

Onda quadrada com proporção
em nível alto/baixo variável

Duty Cycle
10%



Duty Cycle
60%

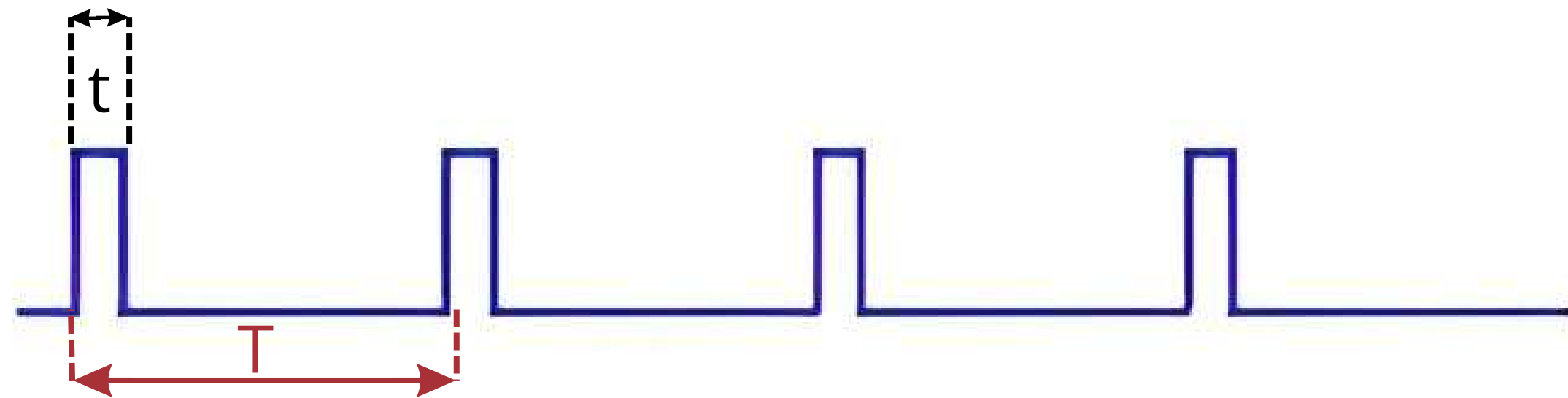


Pulse Width Modulation - PWM

Onda retangular com proporção em nível alto/baixo variável

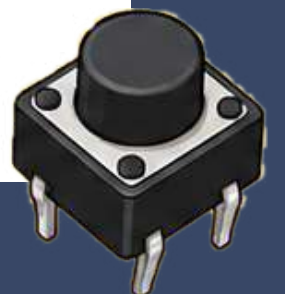
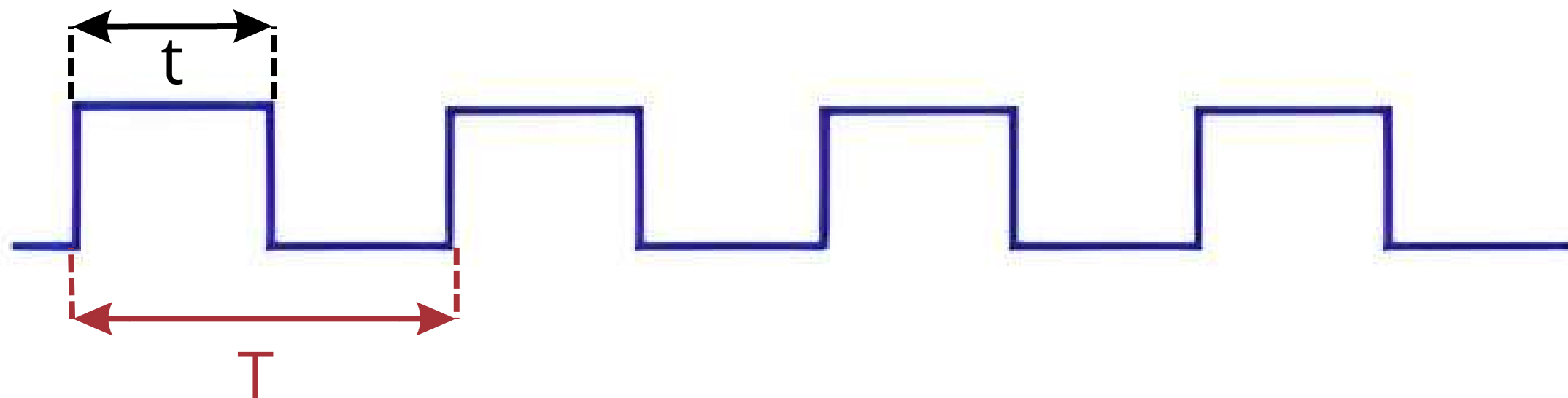
Duty Cycle

$$t/T = 10\%$$



Duty Cycle

$$t/T = 60\%$$



Contador de 0 até 1M 250 mil



Início

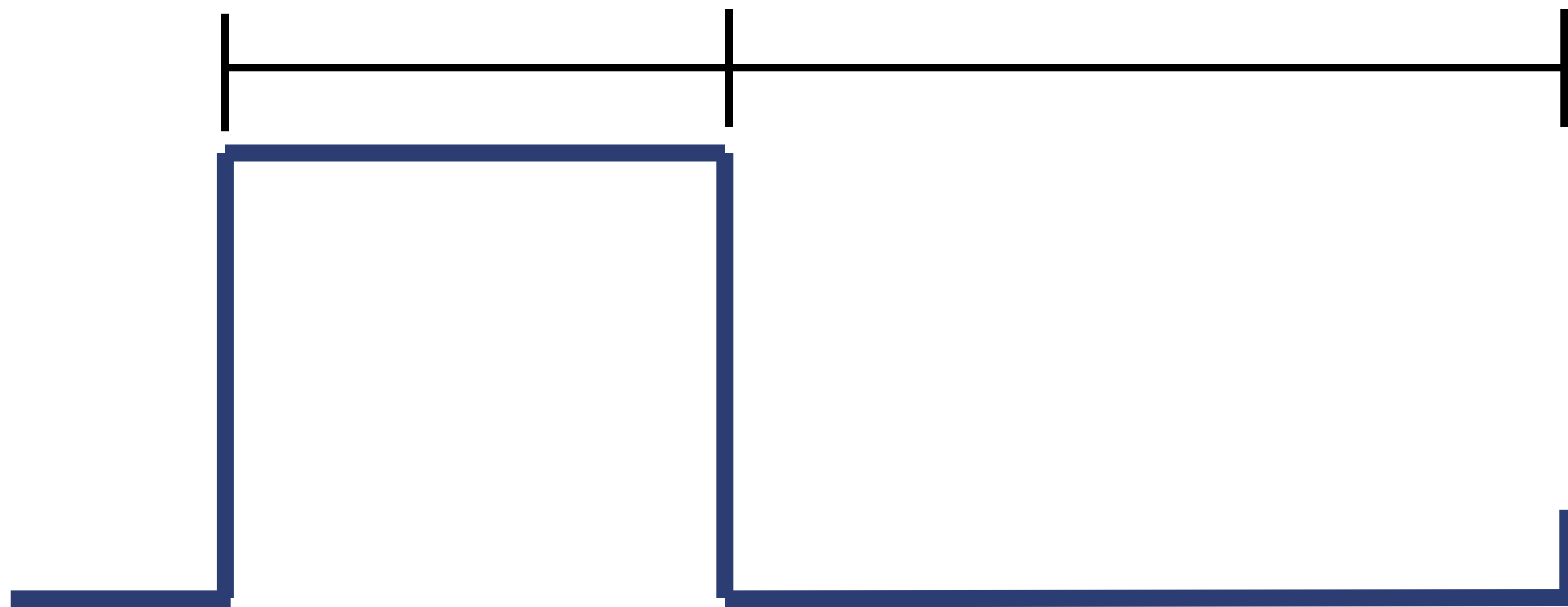
T_{alto}

Fim/reset

0

625 000

1 250 000



...



Contador de 0 até 1M 250 mil



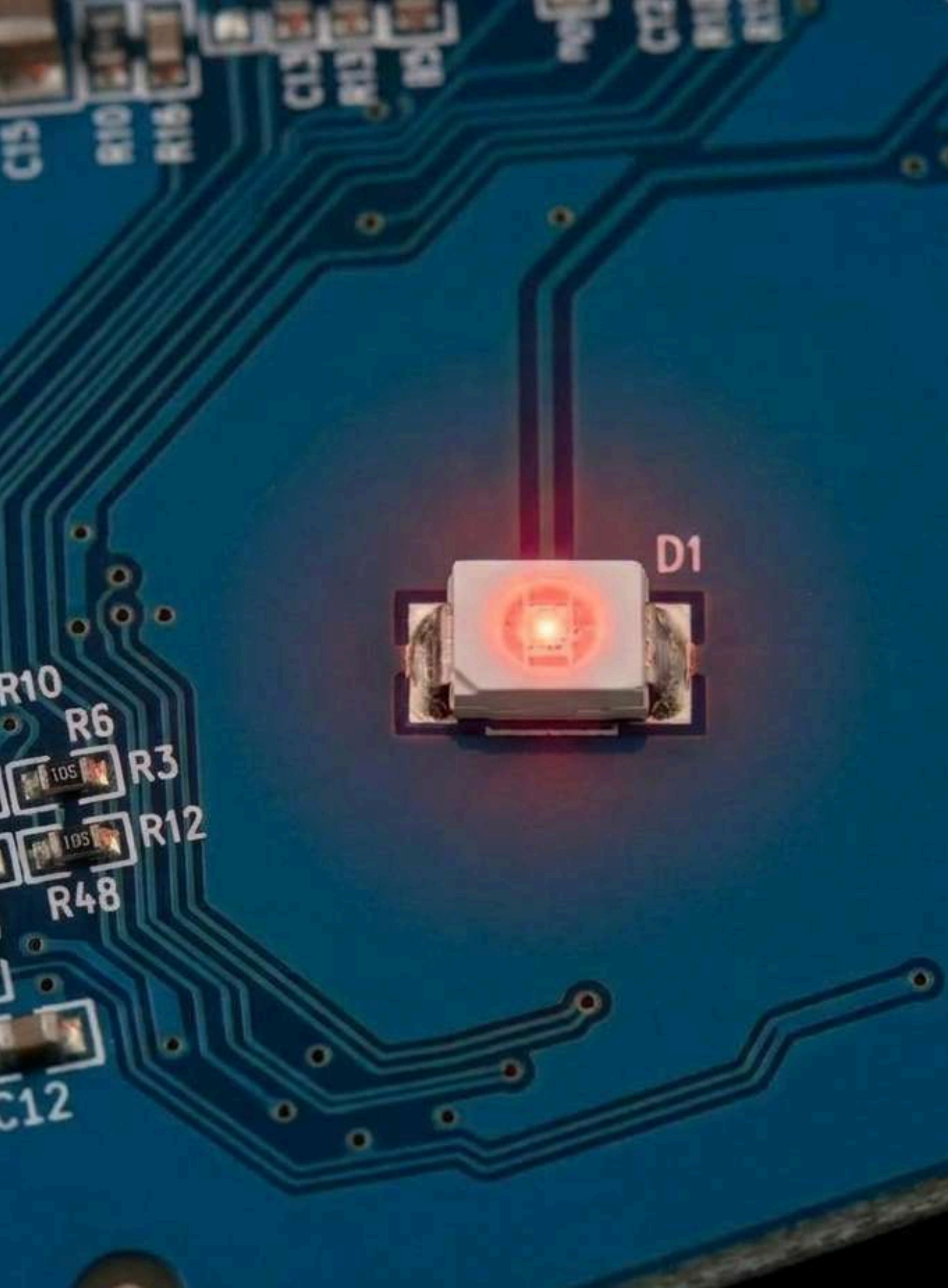
- Relógio de 50 MHz (fixo, cristal)
- Cada pulso tem $1 \text{ s} / 50\,000\,000 = \underline{20 \text{ ns}}$
- $1\,250\,000 * 20 \text{ ns} = \underline{25 \text{ ms}}$
- $1 \text{ s} / 25 \text{ ms} = \underline{40 \text{ ciclos}}$ de PWM por segundo



- **Acompanhe o professor nas alterações de código**



QUARTUS® II



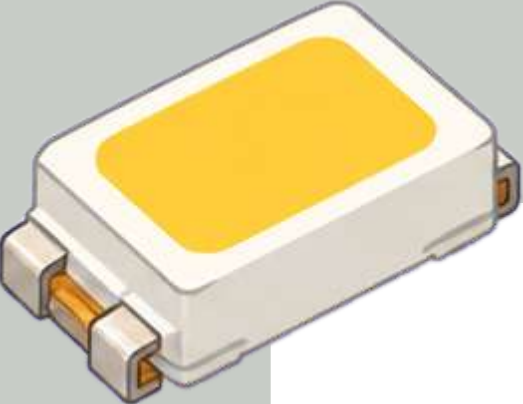
- **Filme o LED com a opção câmera lenta do seu smartphone**

- **Veja o esquemático RTL gerado pelo compilador**
- **Tools →
Netlist Viewers →
RTL Viewer**



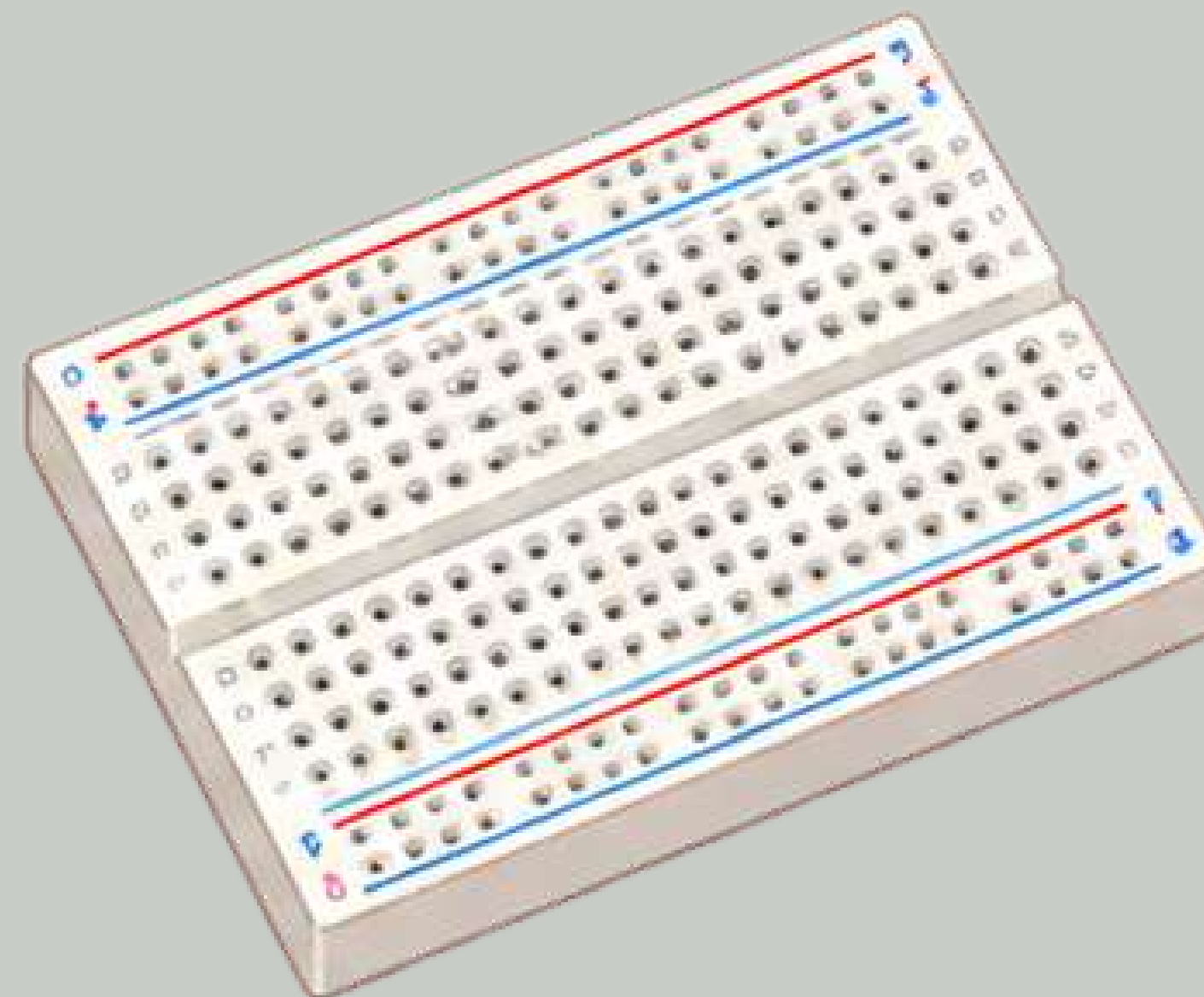
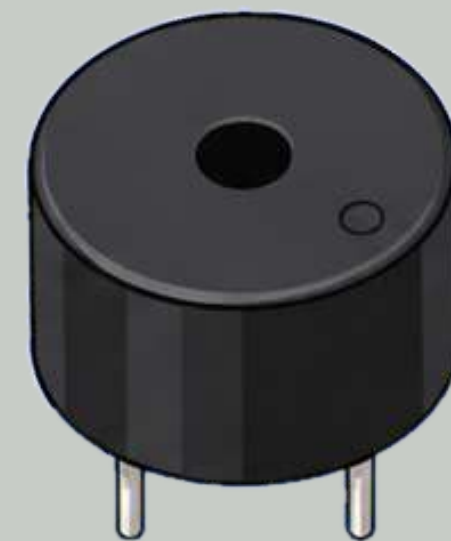
- **Veja a alocação dos CLBs na FPGA**
- **Tools →
Chip Planner**





Projeto final

Código Morse com
buzzer

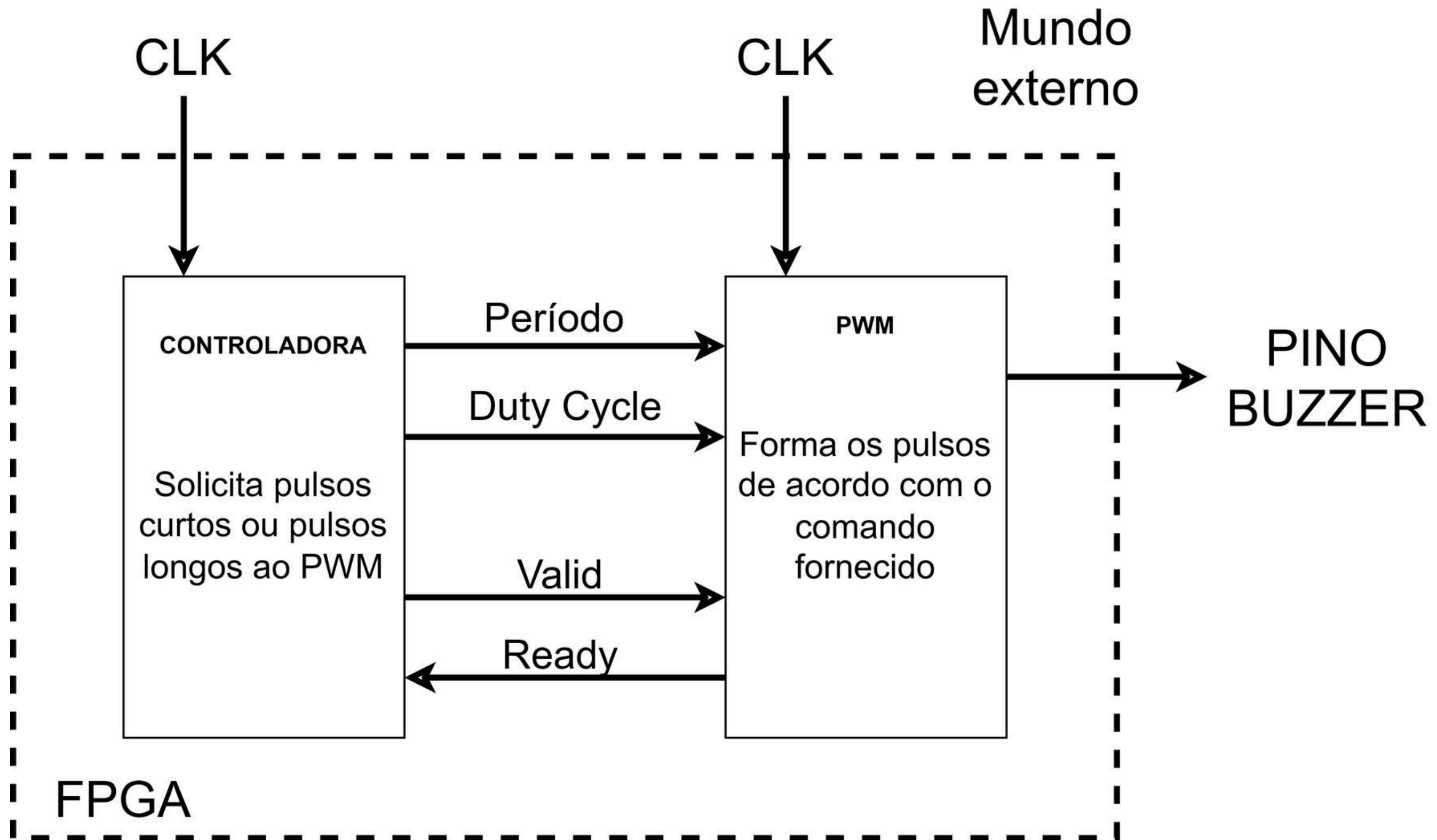


- Abra a pasta
Parte2_codigo_morse
- Abra o projeto
“codigo_morse.qpf”
- Abra o arquivo
“pwm.v”



QUARTUS® II

Controladora e Módulo de PWM



- Acompanhe o professor nas alterações do código



QUARTUS® II

Código Morse

A ● —

B — ● ● ●

C — ● — ●

D — ● ●

E ●

F ● ● — ●

G — — ●

H ● ● ● ●

I ● ●

J ● — — —

K — ● —

L ● — ● ●

M — —

N — ●

O — — —

P ● — — ●

Q — — ● —

R ● — ●

S ● ● ●

T —

U ● ● —

V ● ● ● —

W ● — —

X — ● ● —

Y — ● — —

Z — — ● ●

Onde é possível aplicar esse conhecimento:

cādence

Ferramentas de EDA usadas em chips no seu bolso. **BH**

**invent
VISION**

Forças armadas: monóculo de visão noturna, tanques. **BH**

 **KRYPTUS**
shaping trusted bonds

Criptografia em hardware usada nas urnas eletrônicas. **Campinas**

 **VECTOR**
TRADING

Operações na bolsa de valores aceleradas por hardware. **Remoto**

Disciplina optativa de programação de FPGAs



MINISTÉRIO DA EDUCAÇÃO
CENTRO FEDERAL DE EDUCAÇÃO TECNOLÓGICA DE MINAS
GERAIS
DIRETORIA DE GRADUAÇÃO

Plano de Ensino

CAMPUS DIVINÓPOLIS

DISCIPLINA: Tópicos Especiais em Sistemas de Automação e Hardware: Programação de FPGAs utilizando HDLs

CÓDIGO

Início: **03/2027**

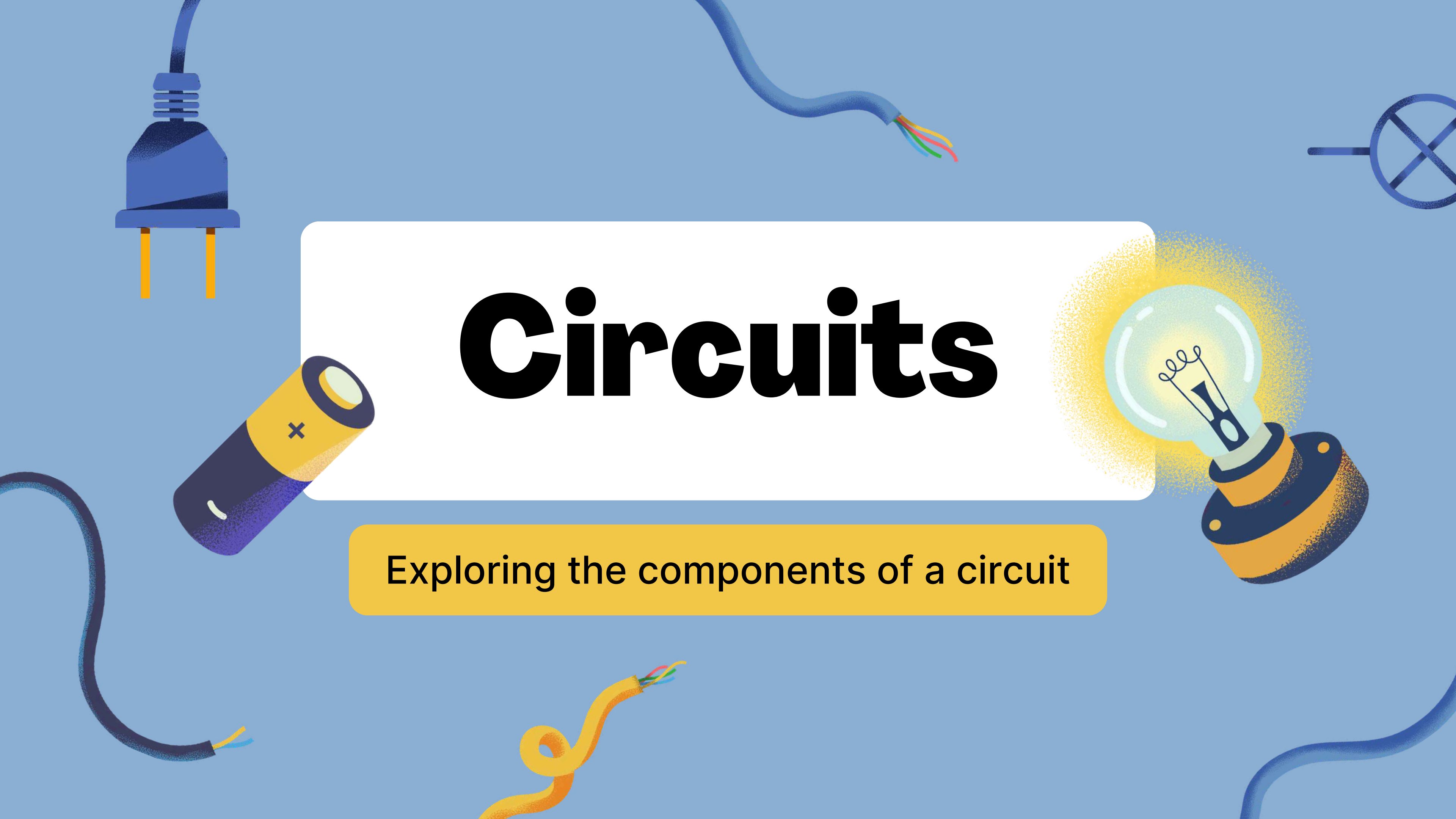
Carga Horária: Total: 60 horas/aula

Semanal: 04 aulas/aula

Créditos: 04

Natureza: Teórico-prática

Área de Formação - DCN: Profissionalizante



Circuits

Exploring the components of a circuit

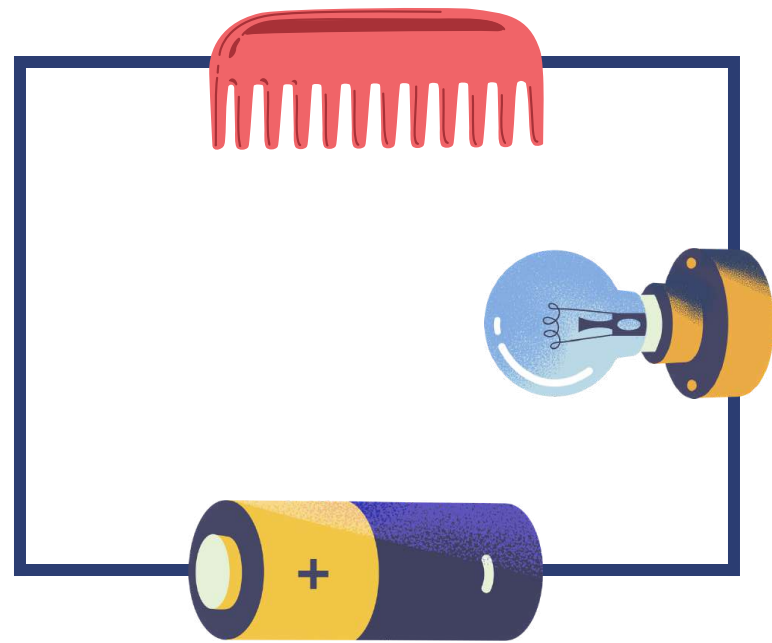


Today's Focus

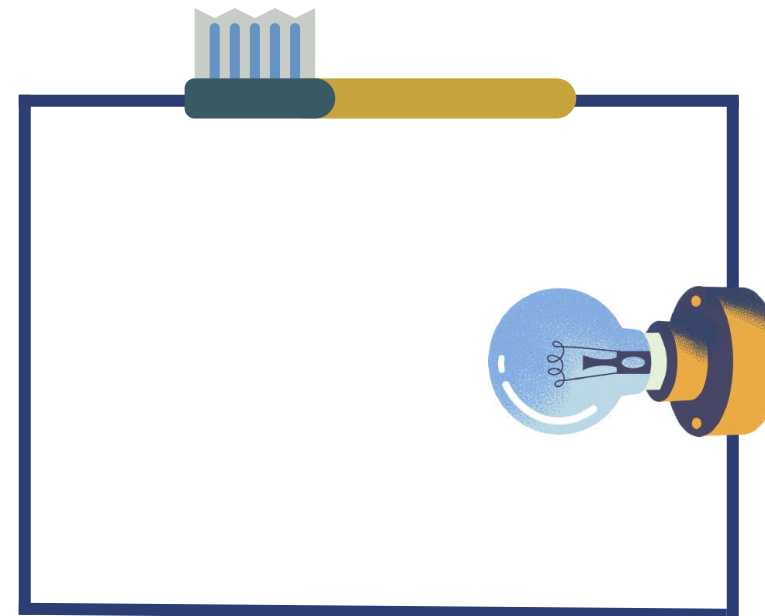
- Electric Circuit
 - Components of Electric Circuit
 - What happens in a circuit?
 - Conductors and Insulators
- 

Circuit or Circus?

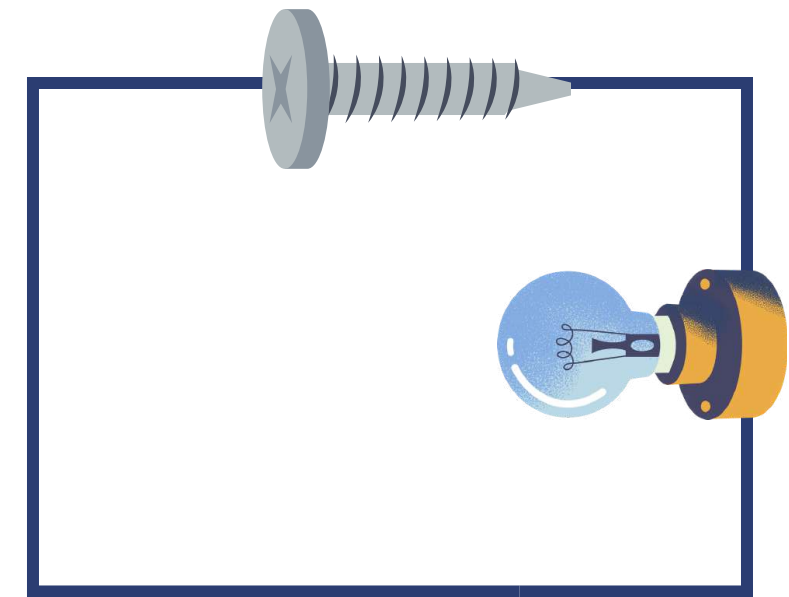
Which of these circuits would work?



A



B

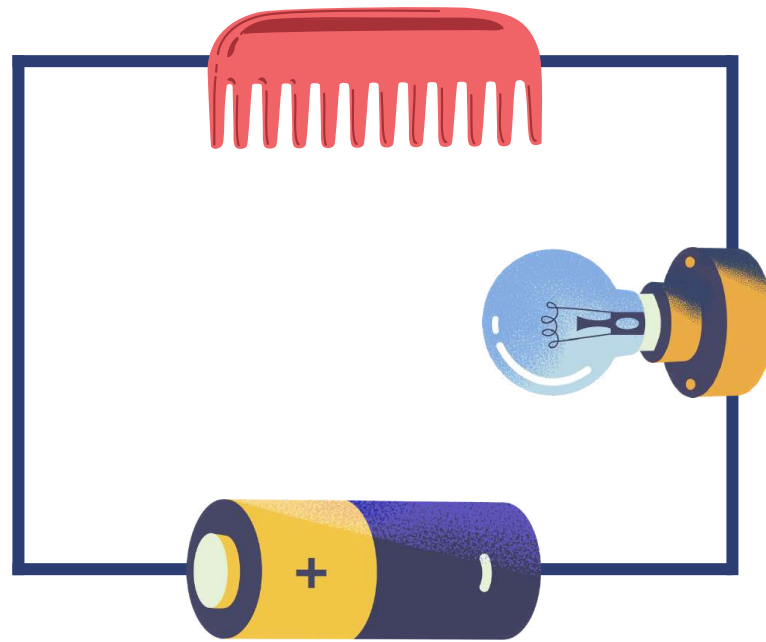


C

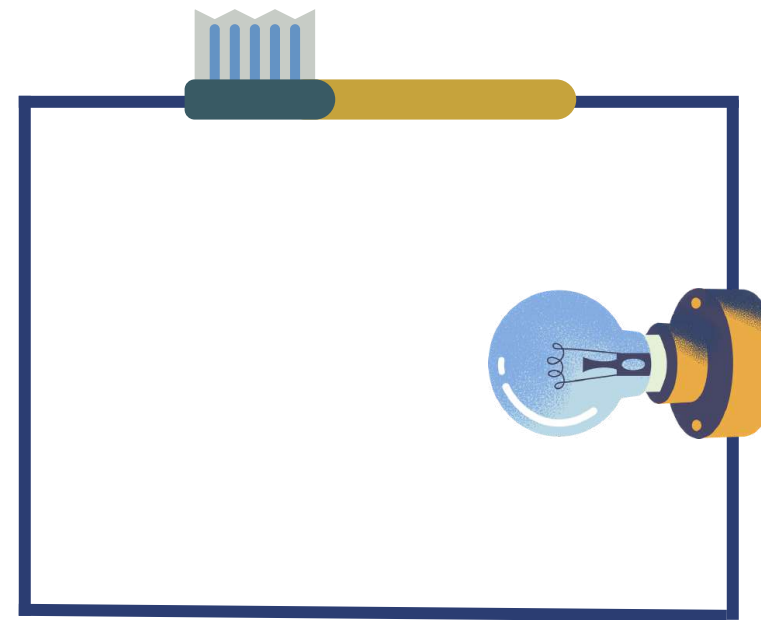
Circuit or Circus?

Which of these circuits would work?

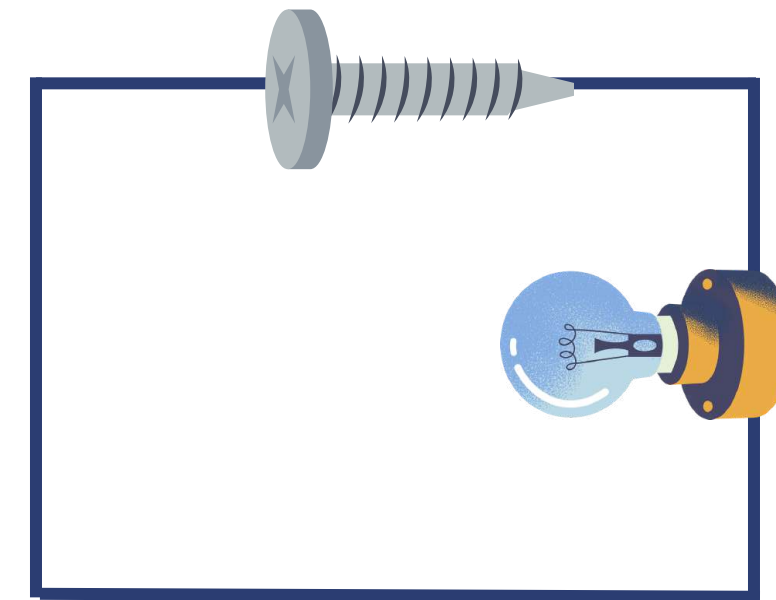
ANSWER
KEY



A plastic comb is not a conductor of electricity.



A plastic toothbrush is not a conductor of electricity, and a power source is missing.



A metal screw is a conductor of electricity, but the power source is missing.

By the end of the lesson, you'll be able to...

Describe what an electric circuit is.

Name at least three (3) components of a circuit.

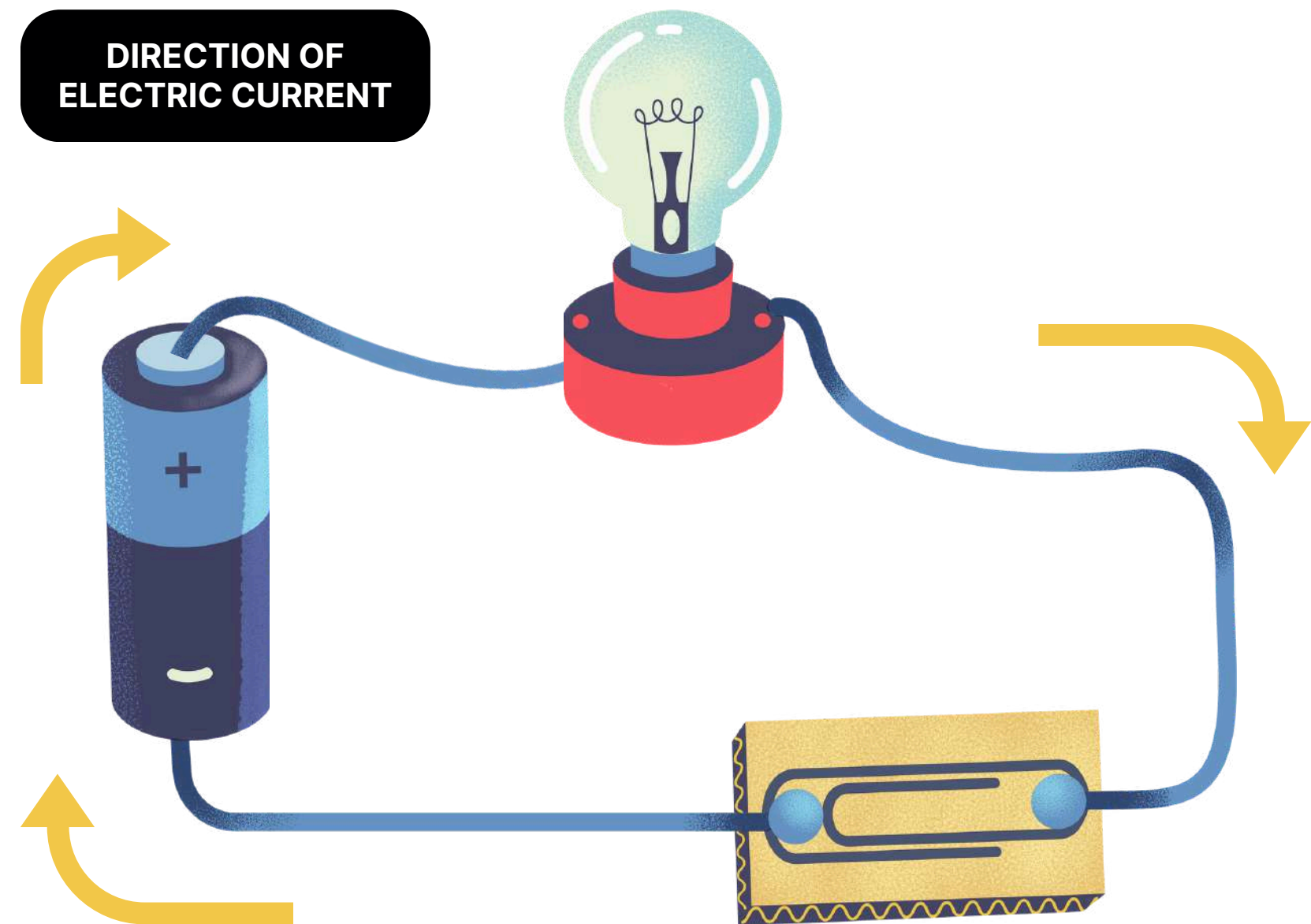
Explain what happens in an electric circuit.

Differentiate conductors from insulators.

What is an electric circuit?

It is a complete path through which electric current flows.

It should be constructed in an unbroken loop.



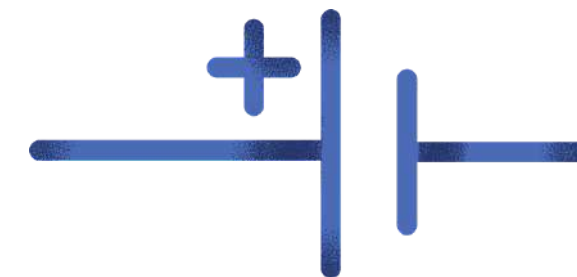
What makes an electric circuit complete?

A simple circuit consists of a battery (or other source), a light bulb (or other load) and conducting wires.

COMPONENT

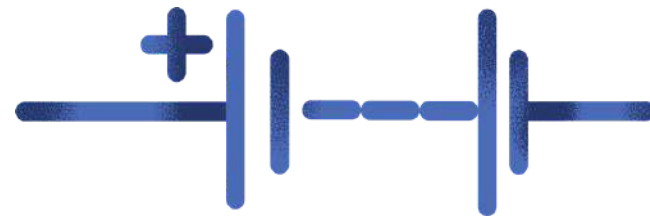


SYMBOL



Other Circuit Components

Shown below are other circuit components and their corresponding electrical symbols:



battery



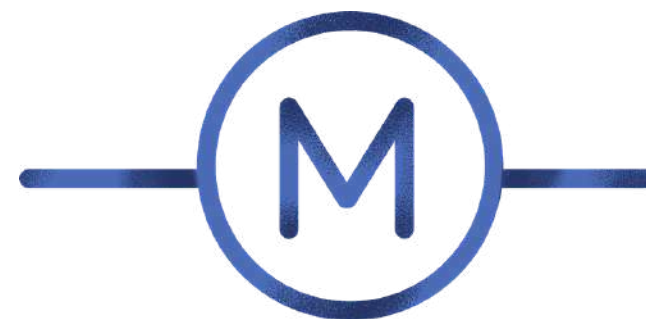
closed switch



open switch



resistor

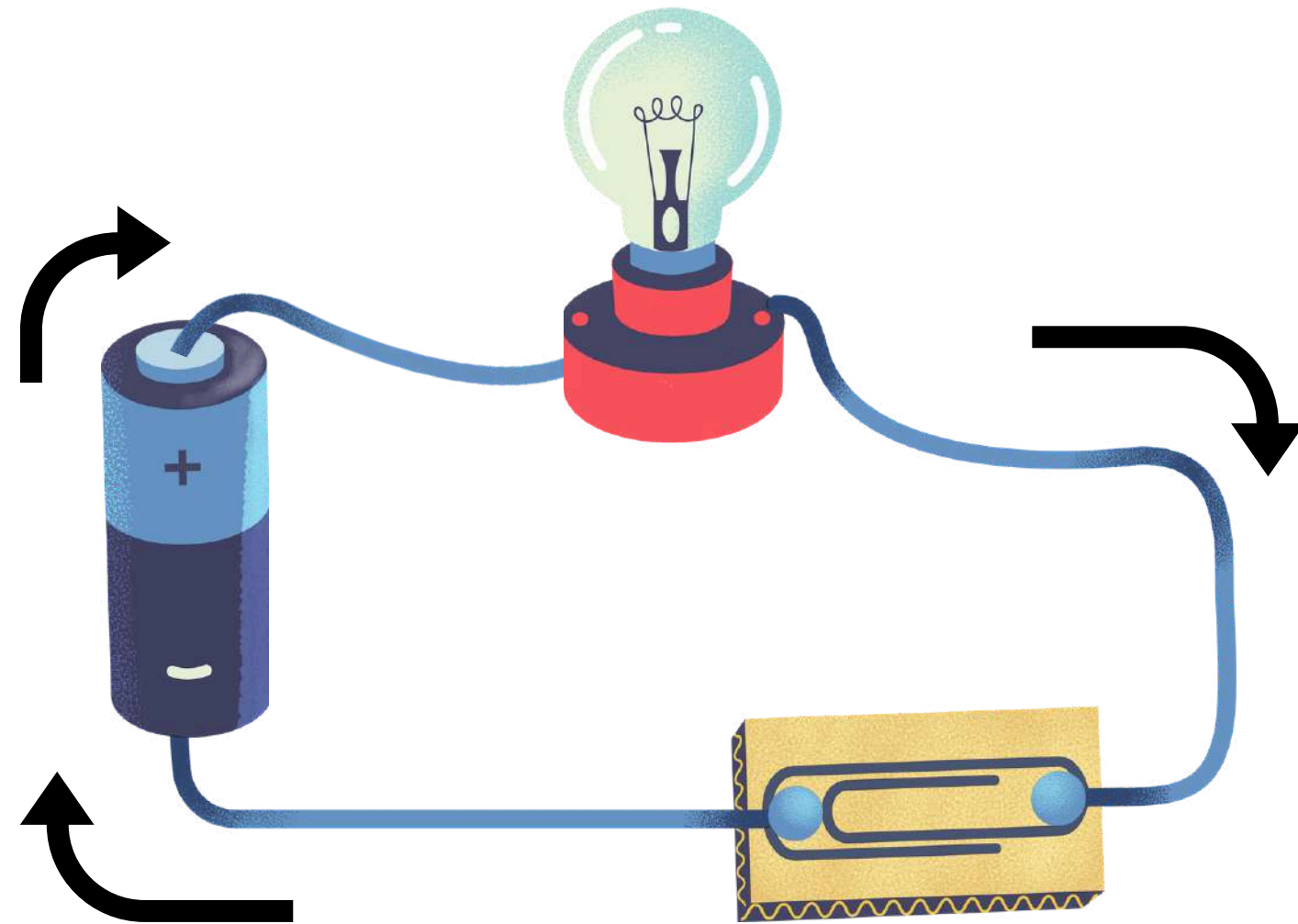


motor



fuse

What happens in a circuit?



A dry cell provides the 'push' needed for electrons to flow in the loop.



These electrons are already present in the wires.



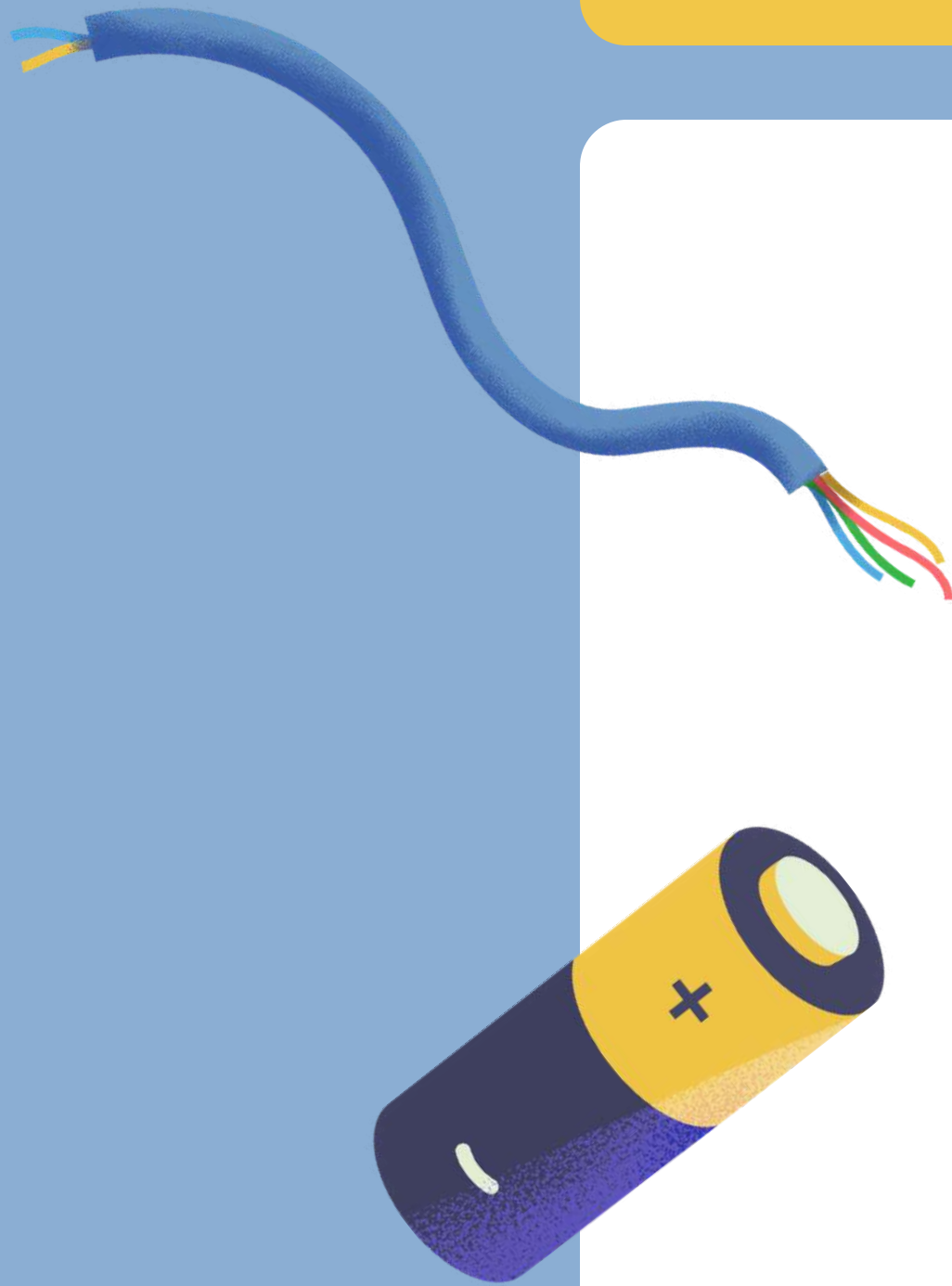
When these electrons pass through a load or a receiver, electrical energy can change into other forms, such as light, heat and sound.

Completing the Circuit

Given a bulb, a wire and a cell,
can you suggest:

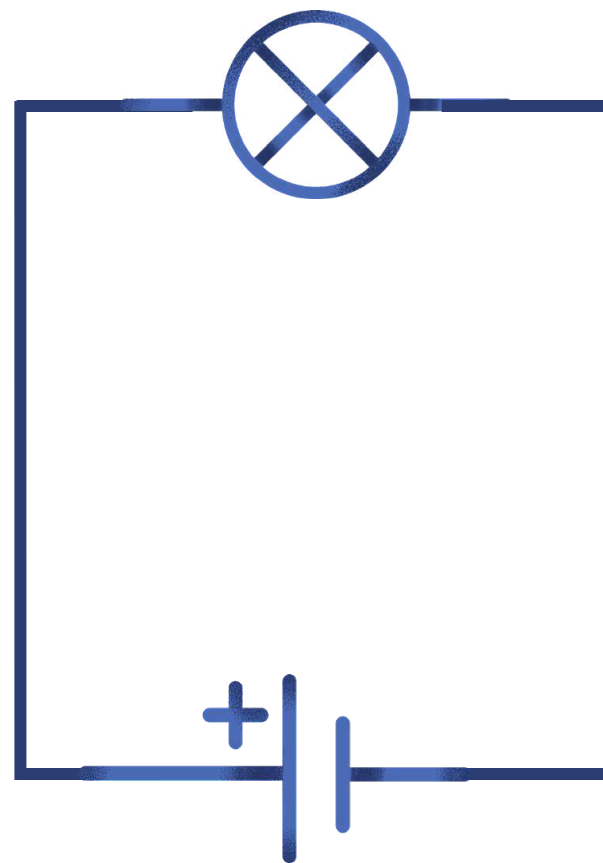
- 1 way to connect them to get the bulb to light up?
- 1 way to connect them such that the bulb does not light up?

Use the electrical symbols when drawing the circuits.

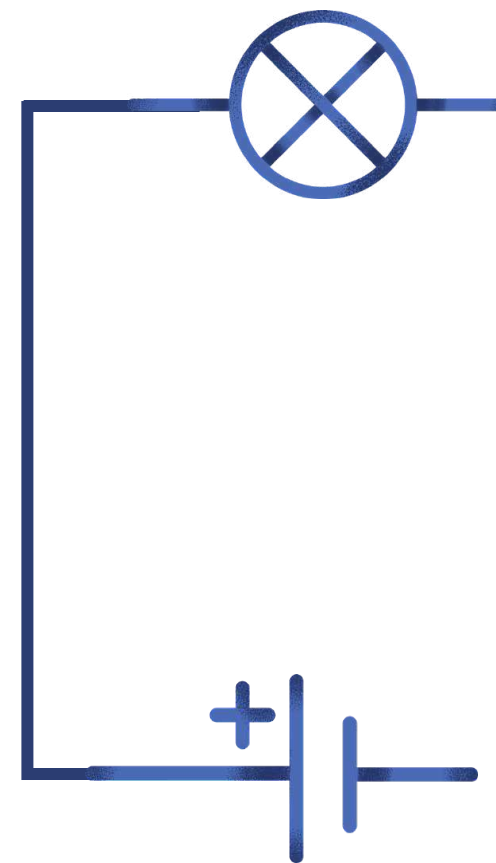


Completing the Circuit

SAMPLE ANSWER



Bulb lights up



Bulb does not light up

Completing the Circuit

EXTENSION ACTIVITY

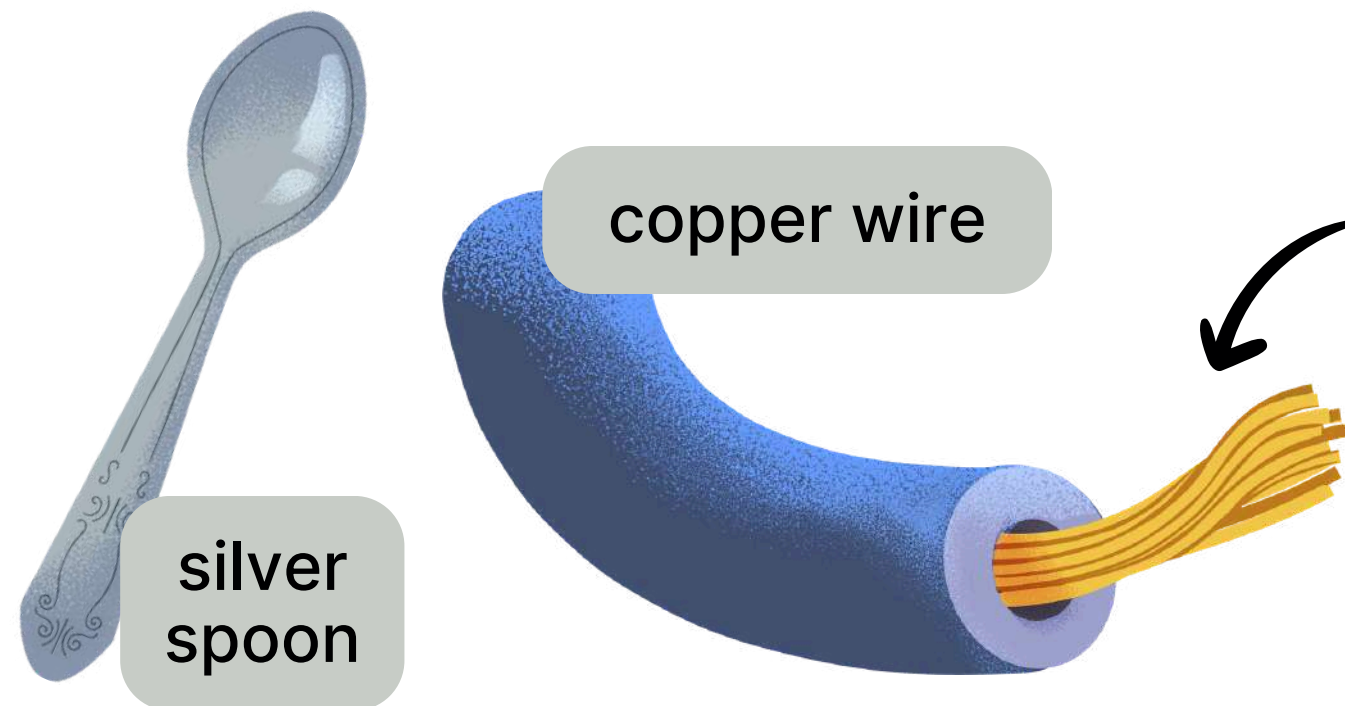


With the help of your teacher or carer, test whether your suggestions and diagrams will make the bulb light using actual batteries, a small lightbulb and wires.

Conductors vs Insulators

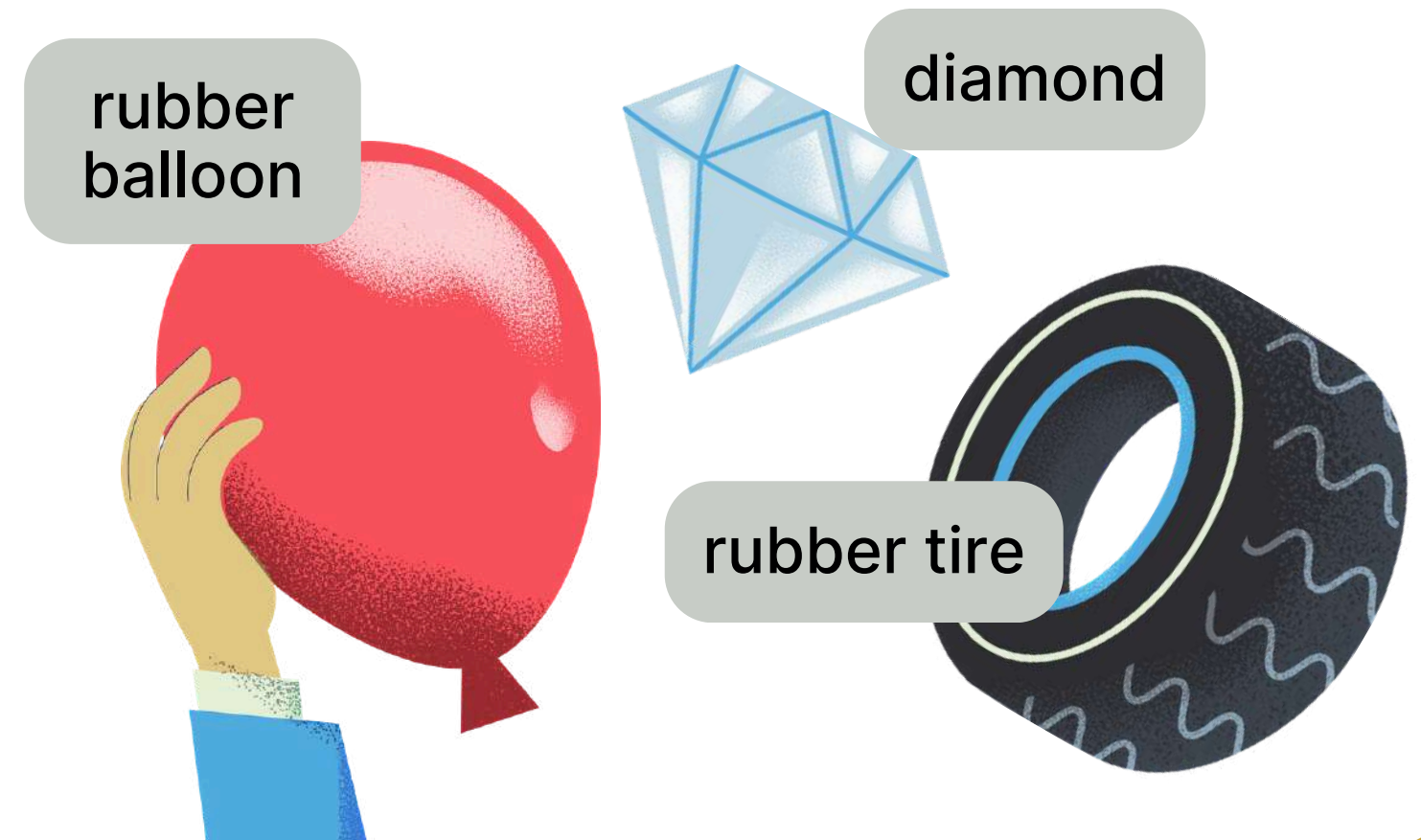
Conductors

These materials allow charges to flow freely.



Insulators

These materials resist the free movement of charges.



Think About It!

Can you name other
conducting materials
aside from copper wire
and silver spoon?



What can electric circuits do?

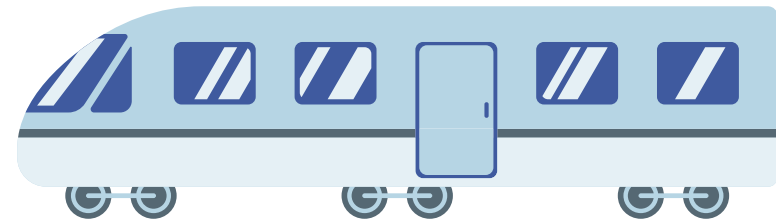
Electric circuits play a crucial role in our daily lives.



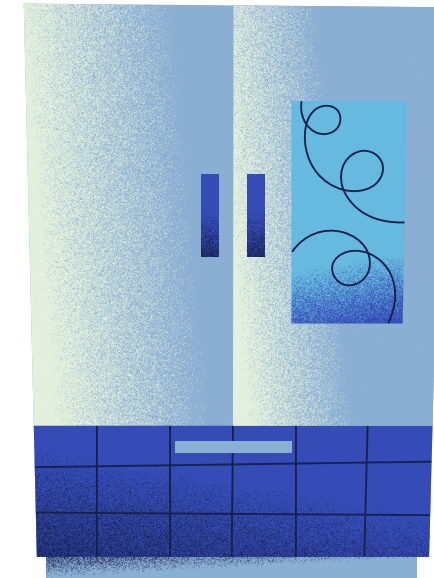
It allows specific devices to produce heat and sound.



It allows light bulbs to light up.



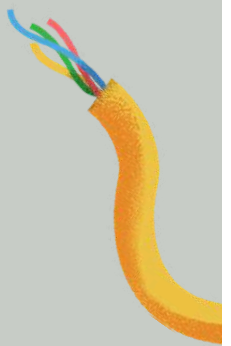
It helps trains move and makes travelling to work or school easier.



It enables the cooling system of a refrigerator to work.

Review

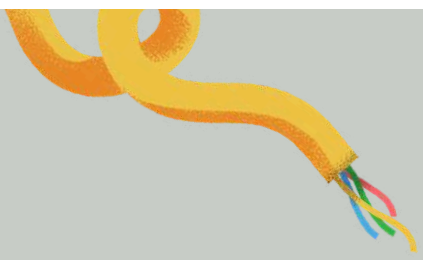
What is an electric circuit?



It is a complete path through which electric current flows.

Some of its components include wires, a bulb and a power source (cell or battery).

A complete circuit features an unbroken loop where charges can flow.



Turning Lemons into Electricity

Find out how a lemon-powered clock works.

Research and share what you've found.

